Cosmology and the Bost–Connes System

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Abstract Starting from the Bost–Connes parent structure and its finite reductions, we develop a common description of the universe as a whole, its dynamical evolution, and observation from within. Joint coherence and correlations connect different observational channels into one whole, while matter and background change one another through formation, maintenance, and exchange. We construct a joint cycle that retains source memory and compensates energy changes, connecting heat exchange, resource restoration, and matter maintenance, and derive an effective temperature law for collective cooling. The phases of loaded matter provide time references, and correlated light signals connect distant events. Under a common calibration, the ratio of the matter clock rate to the light propagation rate determines the cosmological scale. Explicit dynamics yields finite scale growth, spectral redshift, and residual energy in a dark thermal subspace, showing how energy transfer, the persistence of coherence, and continued resource supply affect thermal evolution and spacetime readings.

Keywords Bost–Connes system; cosmology; joint dynamics; thermal evolution; matter clocks; internal observation

Chapter 1. The Universe as a Whole

Dynamics of the Bost–Connes State Space establishes finite reductions and their joint dynamics. *The Unification of Matter, Forces, and Spacetime from the Bost–Connes System* further relates formation, maintenance, and exchange to physical matter and its interactions. [1,2] These results provide a common mathematical basis for the physical description within a given observational scope. Cosmology extends this description to the whole: how are matter and background in different observational scopes connected, and what common structure of states and energies supports them?

We begin with finite reductions. Fixing a prime and a resolution depth specifies the physical content of a finite description; varying the scope of reduction reveals the relations between such descriptions. Tracing these relations back to the Bost–Connes parent structure explains how the channels belong to a single universe.

1.1 Physical Basis of Finite Reduction

Fix an odd prime p and a resolution depth $K \geq 1$. Finite p-adic reduction gives a residue-class space and a complex state space over it. With normalized counting measure,

$$X_{p,K} = \mathbb{Z}/p^K\mathbb{Z},$$
$$\mathcal{H}_{p,K} = L^2(X_{p,K}, \mathbb{C}), \qquad \dim\mathcal{H}_{p,K} = p^K.$$

(1.1)

This space retains zero, nonunits, and units, and describes states by complex amplitudes and their relative phases. The prime p specifies the arithmetic component, and K specifies the resolution depth. [1]

State-space dynamics distinguishes background from directional excitations through valuation shells. The unification theory then relates formation, maintenance, and exchange to specific matter and interactions. [1,2] Denoting the shell-average space by $\mathcal{C}_{p,K}$ and its orthogonal complement by $\mathcal{M}_{p,K}$, we have

$$\boxed{\mathcal{H}_{p,K} = \mathcal{C}_{p,K} \oplus \mathcal{M}_{p,K}.}$$

(1.2)

Matter forms through exchange with the background, and its persistence is supported by actual return and maintenance processes. Once formed, matter can change its internal state and position distribution, affect other matter through exchange, and alter the background through its source action. Matter, background, and the participating auxiliaries thus form an interconnected physical system.

Varying the prime or resolution depth takes us from this finite description to the relations between different observational scopes.

1.2 Algebraic Structure of Observational Channels

Increasing K improves resolution within the same prime component, whereas changing p selects another arithmetic component. The Chinese remainder theorem first places finite coordinates at different primes in a common space. For $p \neq q$ and finite depths K and L,

$$\begin{aligned} M &= p^K q^L, \\ \mathbb{Z}/M\mathbb{Z} &\cong X_{p,K} \times X_{q,L}, \\ L^2(\mathbb{Z}/M\mathbb{Z}) &\cong \mathcal{H}_{p,K} \otimes \mathcal{H}_{q,L}. \end{aligned}$$

(1.3)

A single residue class modulo M determines both finite coordinates, which together reconstruct that residue class. For a joint state ρ_{pq}, tracing out the other factor gives

$$\rho_p = \mathrm{Tr}_q \rho_{pq}, \qquad \rho_q = \mathrm{Tr}_p \rho_{pq}.$$

(1.4)

The statistics at both ends therefore come from the same joint state. Separate readings give the properties of each component, while joint readings also distinguish coherence and correlations between them. In general, ρ_p, ρ_q alone do not determine ρ_{pq}.

These finite relations arise from the full phase decomposition:

$$\mathbb{Q}/\mathbb{Z} = \bigoplus_q \mathbb{Z}[1/q]/\mathbb{Z}, \qquad \widehat{\mathbb{Z}} \cong \prod_q \mathbb{Z}_q.$$

(1.5)

Rational phases decompose into the prime-power components of their denominators, while the dual space retains the p-adic coordinates at every prime. The corresponding finite phase readings are obtained by coordinate projection:

$$\pi_{p,K} : \widehat{\mathbb{Z}} \to \mathbb{Z}_p \to X_{p,K}, \qquad \iota_{p,K}(f) = f \circ \pi_{p,K}.$$

(1.6)

Here f is a function on the finite residue classes, and $\iota_{p,K}(f)$ is the corresponding observable in the parent phase structure. [1] This projection relates finite phase statistics to the parent structure. Distinguishing off-diagonal coherence in the full quantum state requires operations beyond phases alone.

Scaling provides precisely such an operation. In the standard BC representation with orthonormal basis ε_m, phase and scaling act as $e(r)\varepsilon_m = e^{2\pi imr}\varepsilon_m$ and $\mu_p\varepsilon_m = \varepsilon_{pm}$, respectively. Consider

$$\chi_\pm = \frac{\varepsilon_1 \pm \varepsilon_p}{\sqrt{2}}.$$

(1.7)

For every rational r, these states have the same phase expectation but different scaling coherence:

$$\begin{aligned}\langle \chi_\pm, e(r)\chi_\pm \rangle &= \frac{e^{2\pi ir} + e^{2\pi ipr}}{2}, \\ \langle \chi_\pm, (\mu_p + \mu_p^*)\chi_\pm \rangle &= \pm 1.\end{aligned}$$

(1.8)

Their difference lies in the coherence resolved by scaling. The observational scope at a fixed p therefore comprises phases, scaling, and their adjoints, with the associated subalgebra

$$\mathfrak{A}_p = C^*\Big(1, \mu_p, \{e(a/p^K) : a \in \mathbb{Z},\ K \geq 1\}\Big).$$

(1.9)

We use $\mathfrak{A}_p$ to describe a native observational channel. It contains finite combinations of the listed operators and their norm limits; phase statistics at a fixed depth correspond to its phase subalgebra. Physical readings are implemented through the corresponding preparation and operations.

Different channels are further connected by the phase–scaling relation:

$$\boxed{\mu_q^* e(a/p^K)\mu_q = e(qa/p^K).}$$

(1.10)

Scaling in one prime direction can change the phase in another. For $p \neq q$, choosing a depth such that $q \not\equiv 1 \pmod{p^K}$ and taking $a = 1$ gives distinct phase actions. The algebras $\mathfrak{A}_p$ therefore do not generally commute with one another. The Chinese remainder theorem describes the decomposition of finite coordinates, while the native algebras, including scaling, also retain cross-channel actions and their order.

The channels thus share an algebraic origin. The same matter may respond differently in different channels; these responses and their joint statistics enter the state description of the parent structure together.

1.3 States and Energy Spectrum of the Parent System

Phases and scaling at all primes jointly generate the BC parent algebra, whose semigroup crossed-product structure has an explicit construction. [3,4,5] Every rational phase decomposes into finitely many phases with prime-power denominators, and every positive-integer scaling decomposes into a product of prime scalings. Hence

$$\boxed{\mathcal{A}_{\mathrm{BC}} = C^{*}\left(\bigcup_{p\ \mathrm{prime}} \mathfrak{A}_{p} \right).}$$

(1.11)

Here p ranges over all primes, including 2; the finite physical constructions remain in the odd-prime range specified above. The parent algebra contains both operators within individual channels and joint products and ordering relations across channels.

At the level of states, the whole is determined by compatible joint statistics. For a finite set S of primes, write $\mathfrak{A}_{S} = C^{*}(\bigcup_{p \in S} \mathfrak{A}_{p})$, and specify a state ω_{S} on it. If these states satisfy

$$\omega_{T}|_{\mathfrak{A}_{S}} = \omega_{S}, \qquad S \subset T,$$

(1.12)

then a single positive normalized functional is defined on the union of the algebras $\mathfrak{A}_{S}$. Compatibility makes the definition independent of the chosen set, while positivity and normalization ensure boundedness. Since this union is dense in the parent algebra, continuous extension uniquely determines the parent state ω.

The state of each channel is obtained by restriction, $\omega_{p} = \omega|_{\mathfrak{A}_{p}}$. The parent state thus describes both the properties at each end and their joint correlations. Reconstruction of the whole here uses the entire compatible family of joint states.

The common state determines statistical relations between observables, while a single logarithmic parent spectrum relates their local energy scales. In the standard positive-energy representation $\ell^{2}(\mathbb{N}^{*})$, let $\{\varepsilon_{m}\}_{m \geq 1}$ be the positive-integer basis. Then

$$\boxed{H_{\log}\varepsilon_{m} = (\log m)\varepsilon_{m}.}$$

(1.13)

The operator $H_{\log}$ gives a dimensionless logarithmic energy. Once a physical unit E_{*} is fixed, the corresponding energy is $E_{*}H_{\log}$. The parent spectrum sets the reference energy levels; interaction and control energy are included in the balance of the joint dynamics.

Unique prime factorization relates the valuations at different primes within a single spectral state:

$$m = \prod_q q^{a_q(m)}, \qquad \log m = \sum_q a_q(m) \log q.$$

(1.14)

Each m has only finitely many nonzero valuations, and its basis vector corresponds to a tensor product of valuation basis vectors at the primes. Thus

$$\ell^2(\mathbb{N}^*) \cong \bigotimes_q^{\varepsilon_0} \ell^2(\mathbb{Z}_{\geq 0}),$$
$$H_{\log} = \sum_q (\log q) N_q.$$

(1.15)

Here $N_q \varepsilon_m = a_q(m) \varepsilon_m$. The tensor product uses the zero-valuation vector ε_0 at each prime as its reference, and the energy sum is initially defined on the dense domain of finite linear combinations of basis vectors. The valuations at all primes jointly determine the energy of a parent spectral state; a general state also contains superpositions and correlations between spectral components.

At a fixed p, adjacent valuation levels are separated by $\log p$. Accordingly, the finite reduction uses shell energies $E_j = j \log p$, retaining its finite shell spaces, degeneracies, and terminal-layer definition. [1] This correspondence places local energy scales within the parent spectrum. Prime valuations describe the arithmetic composition of energy, whereas the position degrees of freedom of matter are carried by specific displacement operations.

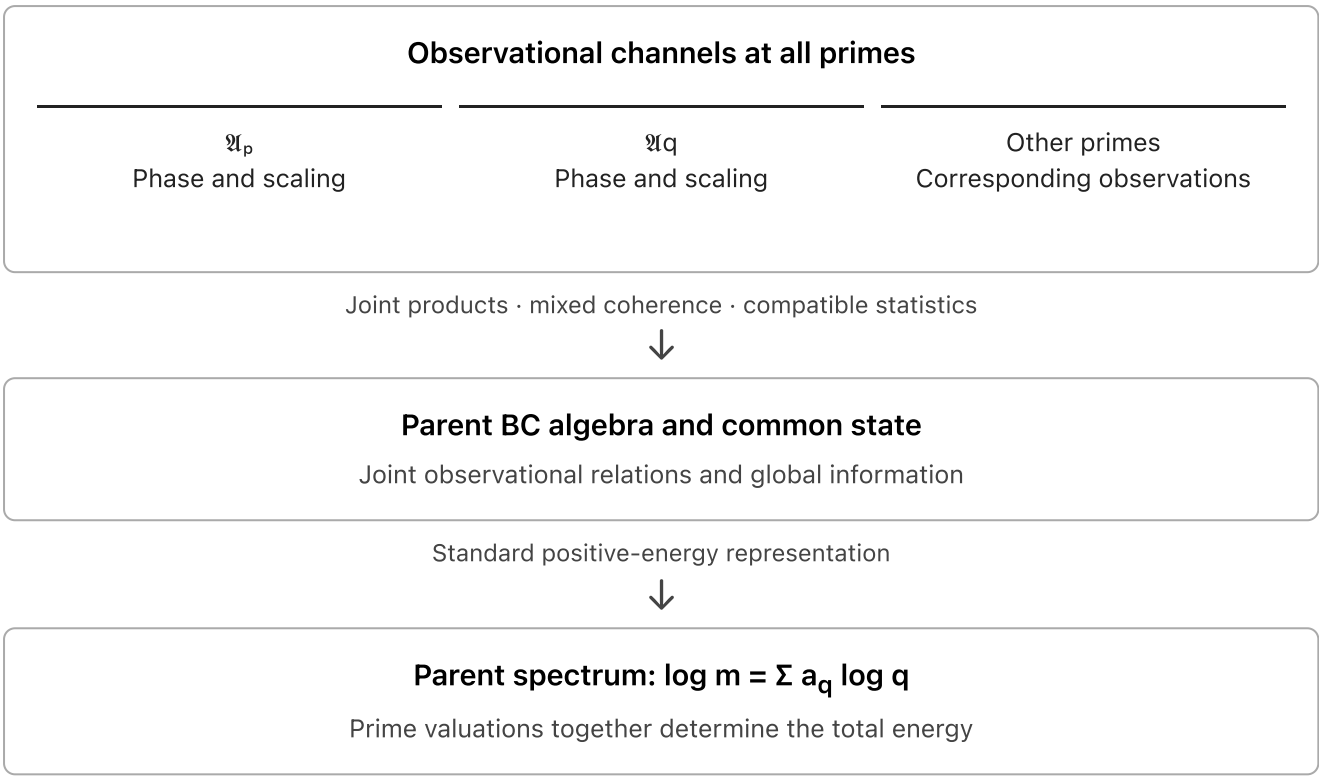


Figure 1-1. Observational channels and the parent structure. Phases and scaling at all primes generate the BC parent algebra, and compatible joint statistics determine the parent state. In the standard positive-energy representation, the parent spectrum decomposes by prime valuations, whose contributions jointly give the total energy of each spectral state.

Prime decomposition of the parent spectrum also determines equilibrium thermal statistics. Let $\beta = E_*/(k_B T)$ be the dimensionless inverse temperature. In the standard Gibbs representation for $\beta > 1$,

$$\tau_\beta = \frac{e^{-\beta H_{\log}}}{Z(\beta)}, \qquad Z(\beta) = \sum_{m=1}^{\infty} m^{-\beta}. \tag{1.16}$$

Summing over valuations at every prime gives

$$\boxed{\begin{aligned} Z(\beta) &= \prod_q \left(\sum_{a=0}^{\infty} q^{-a\beta} \right) \\ &= \prod_q (1 - q^{-\beta})^{-1} = \zeta(\beta). \end{aligned}} \tag{1.17}$$

The Euler product connects the spectral factors at all primes into the full partition function. The equilibrium weight of spectral state m is $m^{-\beta}/\zeta(\beta)$. This thermal structure follows from the common parent spectrum; the change in temperature during actual exchange will be determined by the energy balance.

The trace-normalized thermal state above applies to $\beta > 1$. Equilibrium states for $0 < \beta \leq 1$ are described by KMS states on the BC algebra. [1,3] The parent spectrum and thermal structure provide a common basis for the whole, while a particular state also contains the coherence and correlations generated by its actual processes.

1.4 Unity of the Universe

The state extension in the preceding section connects the whole to local observations: each channel obtains its statistics from the parent state, and the complete compatible family of joint statistics reconstructs that state. Here p labels the observational scope, and K sets the resolution depth.

The information in the whole also includes coherence between channels. This can be calculated directly. For two distinct primes p and q, consider the following states in the standard integer basis:

$$\Psi_{\pm} = \frac{\varepsilon_1 \pm \varepsilon_{pq}}{\sqrt{2}}, \qquad \omega_{\pm}(A) = \langle \Psi_{\pm}, A\Psi_{\pm} \rangle.$$

(1.18)

For any prime r, phase actions in $\mathfrak{A}_r$ leave the integer label unchanged, while scaling and its adjoint change only the r-valuation. All these actions therefore preserve the r-free part $n/r^{v_r(n)}$ of the integer, and norm limits in the generated algebra retain this property. Since 1 and pq have different r-free parts for every r, no $A \in \mathfrak{A}_r$ connects the two basis vectors. These two superposition states therefore have identical single-channel expectations:

$$\begin{aligned} &\omega_{\pm}(A) = \tfrac{1}{2}\big(\langle \varepsilon_1, A\varepsilon_1 \rangle + \langle \varepsilon_{pq}, A\varepsilon_{pq} \rangle\big), \\ &\omega_+|_{\mathfrak{A}_r} = \omega_-|_{\mathfrak{A}_r} \qquad \text{for every prime } r. \end{aligned}$$

(1.19)

Joint scaling, however, maps ε_1 to ε_{pq}. For the self-adjoint observable O_{pq}, direct calculation gives

$$O_{pq} = \mu_p \mu_q + (\mu_p \mu_q)^*, \qquad \boxed{\omega_{\pm}(O_{pq}) = \pm 1.}$$

(1.20)

The two states have identical single-channel statistics and diagonal energy distributions; their difference comes entirely from cross-channel coherence. Joint observation resolves this difference, explaining why state reconstruction requires joint statistics.

Channel operators jointly generate the parent algebra, compatible joint states extend uniquely to the parent state, and the parent spectrum connects all prime valuations to a common energy. The relation between the whole and its observational scopes can be summarized as

$$\boxed{\begin{array}{l}\text{common parent structure and state } (\mathcal{A}_{\mathrm{BC}}, \omega),\\ \omega_p = \omega|_{\mathfrak{A}_p}, \qquad \{\omega_S\}_{S \text{ finite, compatible}} \leftrightarrow \omega.\end{array}} \tag{1.21}$$

On this basis, we take interconnected matter, background, and their common state as the universe as a whole. The parent spectrum provides its energy basis, and the joint state retains both the properties of its parts and their relations. We next specify the structures of these participants, then use their interactions to describe how the universe changes.

Chapter 2. Matter and Background

The Unification of Matter, Forces, and Spacetime from the Bost–Connes System relates formation, maintenance, and interactions in the finite state space to physical matter. [2] The internal structure of matter determines how it can respond, the background state determines the environment of its interactions, and the coherence and correlations they develop continue to affect subsequent processes. This chapter describes the matter carrier, the common background, and their joint state in turn.

2.1 Structure of Matter

Matter has an internal pattern of complex amplitudes. Their magnitudes specify the composition of its components, and their relative phases determine how those components combine in subsequent operations. State-space dynamics characterizes stable modes through pattern preservation under shell-preserving responses, while the unification theory establishes their physical correspondence through actual formation and maintenance. [1,2] At a regular spectral point s_*, the corresponding mode satisfies

$$T_{\mathrm{el}}(s_*)\psi_f = \lambda_f \psi_f, \qquad \lambda_f \neq 0.$$

(2.1)

This spectral relation identifies patterns that preserve their relative structure. The persistence of matter is further realized through actual outflow and return: source, target, and participating auxiliaries jointly carry the same matter, whose internal state changes within this carrier.

In the propagation formulas below, we set $\hbar = 1$ unless physical units are explicitly restored; time and generators are expressed in reciprocal units. Using the carrier embedding $\mathcal{J}_f$, the actual joint generator $\widehat{H_f}$, and the internal generator h_f from Chapter 2 of the unification theory, we have

$$\boxed{\begin{aligned} &\widehat{H_f}\mathcal{J}_f = \mathcal{J}_f h_f, \\ &e^{-is\widehat{H_f}}\mathcal{J}_f = \mathcal{J}_f e^{-ish_f}. \end{aligned}}$$

(2.2)

The equality states that the evolved joint state remains within the same carrier while its logical state changes. This logical state includes internal modes, motion orientation, and position amplitudes; components at different positions jointly constitute a single matter state. [2]

Cosmic evolution requires us to track the structure, state, and persistence of this matter together. Carrier-preserving operations support internal evolution, whereas exchange with other modes changes the population of the original matter carrier. Portions transferred to the background or to products

remain part of the whole; matter of the same kind formed later has its own formation process. Changes in matter composition follow from the succession of these actual operations.

2.2 Joint Background

Matter is identified relative to a background, which formation and return processes themselves change. Within each finite system, state-space dynamics first establishes this reference through the average components of valuation shells. [1] For $0 \le j < K$, write $S_j = \{x \in X_{p,K} : v_p(x) = j\}$; the terminal shell is $S_K = \{0\}$, where v_p is the p-adic valuation. Denote the normalized average state of each shell by e_j. Then

$$\begin{aligned} &\mathcal{C}_{p,K} = \operatorname{span}\{e_0, \ldots, e_K\}, \qquad \mathcal{M}_{p,K} = \mathcal{C}_{p,K}^{\perp}, \\ &\Pi_p + P_p = I. \end{aligned}$$

(2.3)

Averaging over the unit group gives Π_p. It retains amplitudes invariant under the unit action within each shell, whereas P_p retains the nontrivial directions. Background and directions are orthogonal components of the same system and may still possess coherence between them.

The normalized constant function $b_p = 1$ provides a common reference across shells. With $h_{p,j} = |S_j|/p^K$, we obtain

$$b_p = \sum_{j=0}^{K} \sqrt{h_{p,j}}\, e_j, \qquad \sigma_{b,p}^{\mathrm{sh}} = \sum_{j=0}^{K} h_{p,j} P_{e_j}.$$

(2.4)

The former retains intershell phases; the latter is a preparation dephased in shell energy. Which state is supplied by restoration determines the coherence available to subsequent operations.

References at different primes can be identified exactly in a common finite space. For $p \neq q$, set $M = p^K q^L$ and define, through the Chinese remainder theorem,

$$\begin{aligned} (J_p f)(x_p, x_q) &= f(x_p), \\ (J_q g)(x_p, x_q) &= g(x_q). \end{aligned}$$

(2.5)

Both maps are isometric embeddings when the spaces use normalized counting measure. The constant functions give the same vector in the common space:

$$\boxed{J_p b_p = J_q b_q = b_M = b_p \otimes b_q.}$$

(2.6)

The overlap of the two embeddings further satisfies

$$\begin{aligned} \langle J_p f, J_q g\rangle &= \langle f, b_p\rangle\langle b_q, g\rangle, \\ J_p^{\dagger} J_q &= |b_p\rangle\langle b_q|, \\ J_p\mathcal{H}_p \cap J_q\mathcal{H}_q &= \mathbb{C} b_M. \end{aligned}$$

(2.7)

The final equality follows because a function lying in both images must depend only on x_p and only on x_q: fixing either coordinate forces it to be constant. Local references thus belong to one common background, while their deviations orthogonal to the constant function retain distinct directional identities.

Let $\mathcal{D}_p, \mathcal{D}_q$ denote dephasing in the shell energies on the two sides. Then

$$(\mathcal{D}_p \otimes \mathcal{D}_q)(P_{b_M}) = \sigma^{\mathrm{sh}}_{b,p} \otimes \sigma^{\mathrm{sh}}_{b,q}.$$

(2.8)

This gives a joint background compatible with the local preparations. The common reference fixes the relation between descriptions; whether matter shares contact with a nontrivial background degree of freedom is determined by the support of the actual interaction.

Background degrees of freedom participating in maintenance, heat exchange, and restoration have different dynamical roles. The shell-average component within the system receives exchanges from the directional sector; the thermal environment carries the corresponding changes of energy and charge; maintenance resources determine the currently available amplitude of the operation; and return auxiliaries carry coherence through the full process and return it. All these participants belong to the universe, so the environment of local matter is itself internal to the whole.

For a specified set of participants, denote the actual state of the common background by $\rho_{\mathcal{B}}$. Its occupations, coherence, and correlations between components are contained in this state and in its joint state with matter. Even backgrounds with equal mean energy may affect subsequent matter differently because of coherence or contact correlations. Constraints on coherence conversion beyond the energy distribution are also studied in the resource theory of quantum thermodynamics. [6]

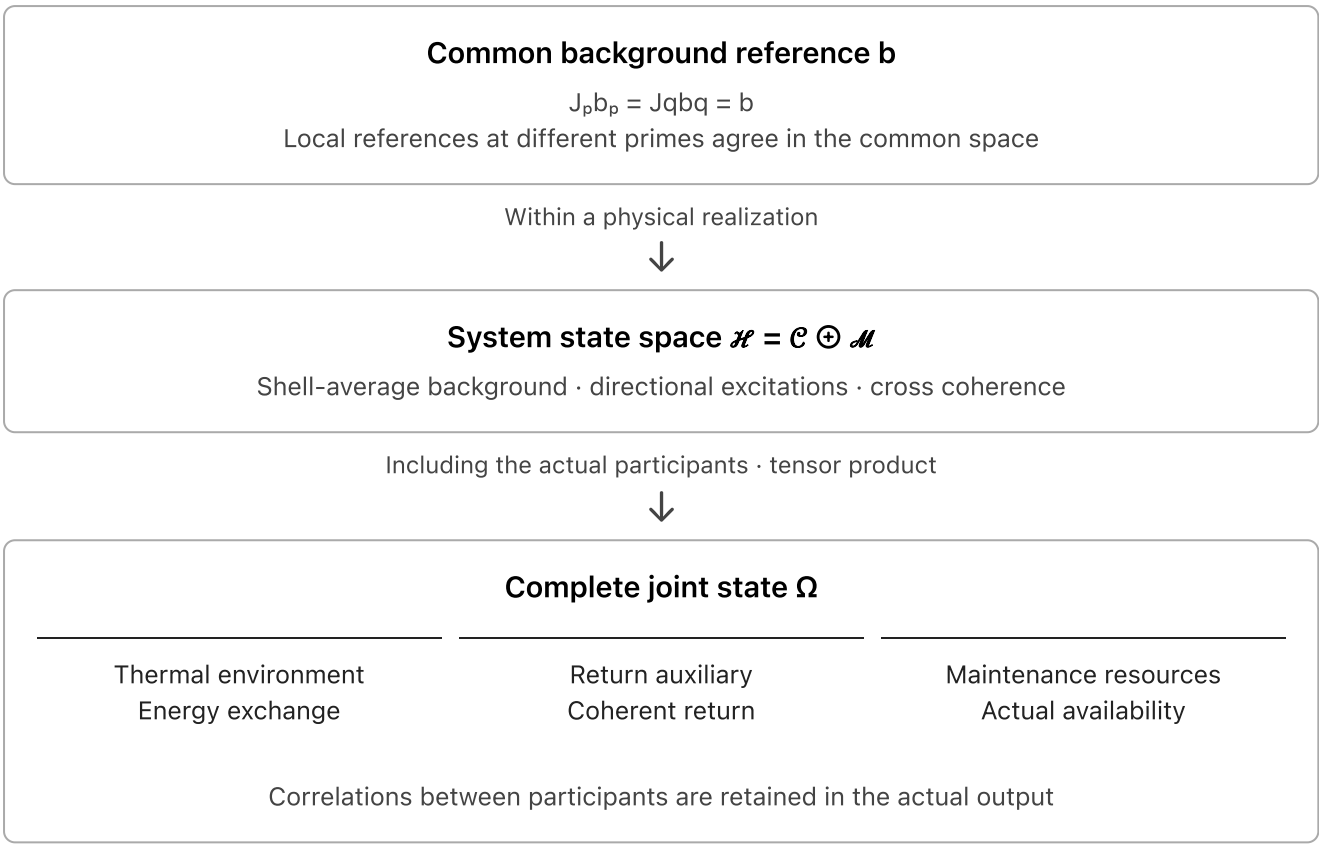


Figure 2-1. Background structure and actual participants. A direct sum distinguishes the shell-average background from directional excitations within a single system; the thermal environment, return auxiliaries, and maintenance resources form a joint space in the actual process. Background references at different primes are related by a common finite representation, while specific shared actions are determined by the same participants coupled into the process.

2.3 Joint State of the Whole

The relation between matter and background is first retained within the same system state. With the background and direction projections Π, P, any density operator decomposes as

$$\boxed{\rho = \Pi\rho\Pi + P\rho P + \Pi\rho P + P\rho\Pi.}$$

(2.9)

The first two terms describe the composition of each part; the last two retain their coherence. Exchange can respond to both these populations and their phase relations, so two independent conditional states generally cannot replace the full ρ. [1]

The thermal environment, return auxiliaries, and other actual participants further form a joint space. For example,

$$\mathcal{K} = \mathcal{H}_S \otimes \mathcal{E}_T \otimes \mathcal{E}_R \otimes \mathcal{E}_{\text{rest}},$$
$$\Omega \geq 0, \qquad \text{Tr}\Omega = 1.$$

(2.10)

The system $\mathcal{H}_S$ retains its internal decomposition $\mathcal{C} \oplus \mathcal{M}$, while the tensor factors represent physically distinct participants. Their marginal states describe individual properties, and the joint state also retains correlations created by their interactions. Mutual information and conditional entropy characterize this structure, with strong subadditivity of joint entropy constraining the relations between them. [7] After an exchange, the output on one side is generally correlated with the other; subsequent operations involving these participants remain determined by that joint state.

To pass from the parent structure to an actual process, we use a common realization with specified participants, energy structure, and connections. In a representation space of the parent algebra that accommodates this finite process, let π be the representation and Ω the embedded joint density operator. Then

$$\omega(A) = \text{Tr}[\Omega\pi(A)], \qquad \omega_p = \omega|_{\mathfrak{A}_p}.$$

(2.11)

The specific preparation and connections determine the actual participants, and each channel obtains its properties from the same joint state. Every operation produces a new joint state of matter and background, through which cosmic evolution continues.

Chapter 3. Evolution of the Universe

Exchange changes both matter and background, and the altered background in turn affects the formation and maintenance of matter. Beginning with a single exchange, this chapter connects maintenance, motion, and feedback into a complete process, then derives thermal evolution from its energy balance. The operations used here follow the processes established in state-space dynamics and the unification theory. [1,2]

3.1 Exchange between Matter and Background

Chapter 2 distinguished matter directions from the shell-average background within the same system. During exchange, the background can supply amplitude to a matter direction, and the matter direction can return amplitude to the background; the jointly participating environment carries the corresponding energy difference. State-space dynamics constructs this operation using joint endpoints matched in energy and charge. [1] Take a system matter direction $|\nu\rangle$, a shell-average background state $|e_a\rangle$, and two environment states $|u\rangle$ and $|\upsilon\rangle$, with energies E_ν, E_a and $\epsilon_u, \epsilon_\upsilon$, respectively. If

$$E_a - E_\nu = \epsilon_u - \epsilon_\upsilon,$$

(3.1)

then $|\nu, u\rangle$ and $|e_a, \upsilon\rangle$ have equal joint bare energy. When their charges also match, define

$$L = |e_a, \upsilon\rangle\langle\nu, u|, \qquad H_{\text{ex}} = i\kappa_{\text{ex}}(L - L^\dagger).$$

(3.2)

Here κ_{ex} is the exchange strength. The self-adjoint operator H_{ex} commutes with the joint bare energy. An interaction time s gives the exchange angle $\alpha = \kappa_{\text{ex}} s$ and propagator

$$\begin{aligned} U_{\text{ex}} &= e^{\alpha(L - L^\dagger)}, \\ U_{\text{ex}}|\nu, u\rangle &= \cos\alpha|\nu, u\rangle + \sin\alpha|e_a, \upsilon\rangle, \\ U_{\text{ex}}|e_a, \upsilon\rangle &= \cos\alpha|e_a, \upsilon\rangle - \sin\alpha|\nu, u\rangle. \end{aligned}$$

(3.3)

The same rotation describes formation and return. Amplitude at the background endpoint can enter the matter direction, and amplitude at the matter endpoint can return to the background; their relative phase contributes to the net change.

This is especially clear for a general correlated input. Compress the actual joint state to the two endpoints ordered as $(|\nu, u\rangle, |e_a, v\rangle)$, denote the resulting, not necessarily normalized, two-dimensional block by σ, and set $p_1 = \sigma_{11}$, $p_2 = \sigma_{22}$. Expanding $U_{\mathrm{ex}}\sigma U_{\mathrm{ex}}^{\dagger}$ gives the change in the matter population:

$$\boxed{\Delta p_1 = \sin^2\alpha\,(p_2 - p_1) - \sin(2\alpha)\mathrm{Re}\sigma_{12}.}$$

(3.4)

The first term comes from the initial population difference between the endpoints, and the second from actual joint coherence. Coherence can drive transfer even when the populations are equal. The input state and the complete operation jointly determine the direction of exchange. Coherence left by one exchange may contribute to the next, so subsequent evolution must retain both output populations and phase relations.

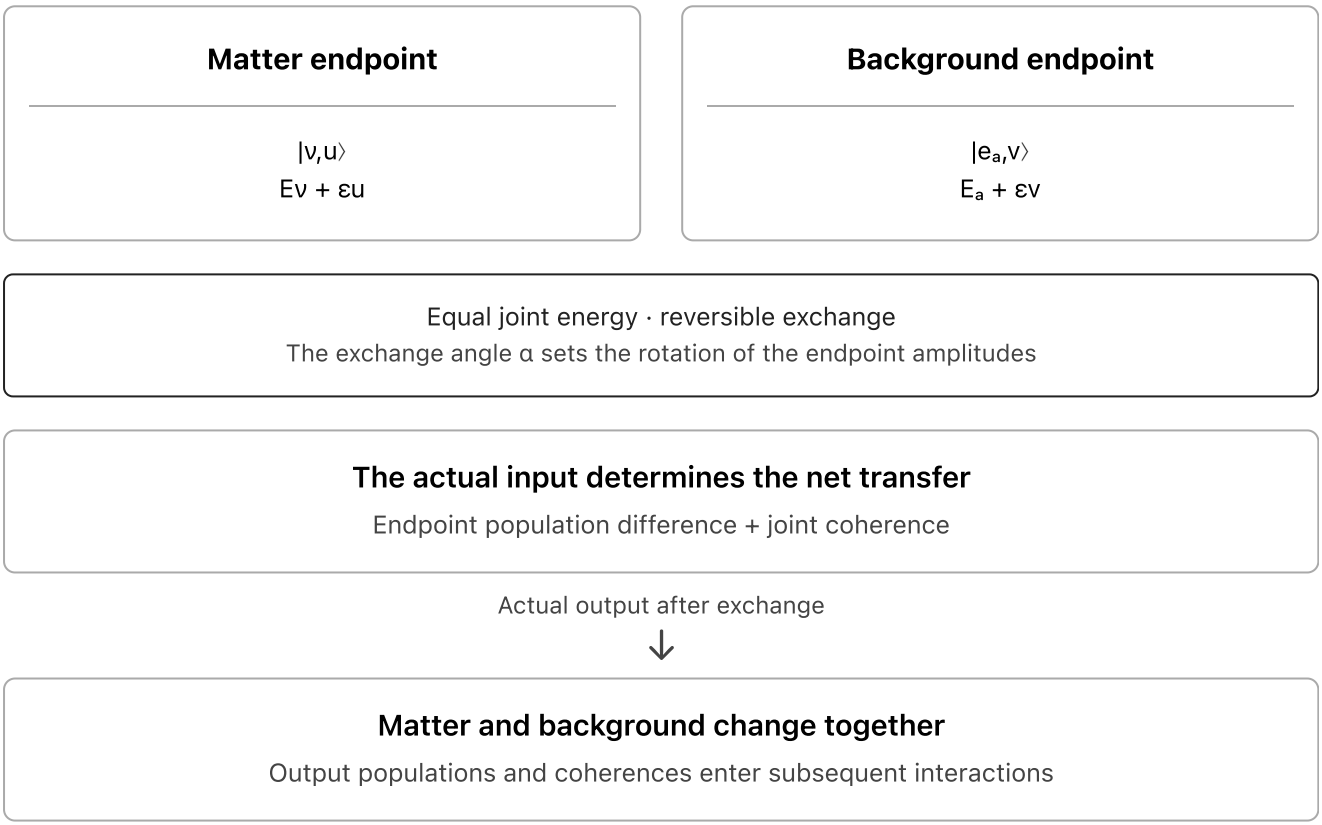


Figure 3-1. Exchange between equal-energy endpoints. One joint configuration of background and environment has the same total energy as another configuration of matter and environment. Unitary exchange changes the endpoint amplitudes, with net transfer depending on both populations and coherence. Matter and background outputs arise from the same operation and jointly enter the subsequent process.

Exchange specifies how matter populations change. Matter that persists after the exchange continues to change its internal state and position distribution, which determine how it participates in later interactions.

3.2 Maintenance and Motion of Matter

The persistent carrier in Chapter 2 accommodates the internal modes, motion orientation, and position amplitudes of the same matter. As long as the carrier is preserved, internal phases and the position distribution can continue to evolve within it. These operations may also occur simultaneously with exchange, their actual combination being determined by the joint generator. Section 2.4.6 of the unification theory establishes shared motion through bidirectional exchange between adjacent positions. When opposite exchanges share a coherent port and the two adjacent endpoints undergo equal, simultaneous maintenance, the logical generator is

$$h_f = M_f \otimes Y \otimes I_x + g I_f \otimes Z_{\mathrm{mov}} \otimes D_x,$$
$$Y^2 = Z_{\mathrm{mov}}^2 = I, \qquad \{Y, Z_{\mathrm{mov}}\} = 0.$$

(3.5)

The operator M_f acts on internal modes and is the mass matrix of the identified matter family; Y and Z_{mov} act on the same motion doublet. Here g is the position-exchange strength, and $D_x = (S^\dagger - S)/(2i)$ connects adjacent positions, with $S|x\rangle = |x+1\rangle$. The first term maintains the internal phase, and the second coherently transfers amplitude between adjacent positions. Position remains a logical degree of freedom in the state space; a unit of length is established by actual measurement.

For a position mode $\langle x|\kappa\rangle \propto e^{i\kappa x}$, we have $D_x \mapsto \sin\kappa$. The spectral relation established from the common generator is [2]

$$E_{f,j,\pm}(\kappa) = \pm\sqrt{\mu_{f,j}^2 + g^2 \sin^2\kappa},$$

(3.6)

where $\mu_{f,j}$ is an eigenvalue of M_f. Internal maintenance and position change thus belong to the evolution of the same matter state. Phase changes modify its superposition relations in the next operation, while transfer of position amplitude changes which participants it may encounter.

Background resources gate the maintenance term through an availability operator B, giving $M_f \otimes Y \otimes B$. Different phase components of matter act differently on the resources, whose changes feed back into subsequent maintenance. This feedback is carried by the joint output of matter and background.

3.3 Feedback between Matter and Background

Feedback begins with the background left after an interaction. Its occupation, phase, and correlations with matter directly determine the response at the next contact. We first calculate how matter writes into the background, then connect this change to matter maintenance.

When matter changes the background, it acts as the source of that change. Section 4.4.4 of the unification theory establishes an explicit process: the complete protected action F_f of matter drives a prepared auxiliary; its positive and negative outputs are routed to the unavailable and available states of a fresh resource, respectively; this resource is then swapped into the background. [2] The background degree of freedom that receives the resource is called a background slot. The operator F_f is the dimensionless representation of the original action within the persistent carrier, including its actual paths and formation populations.

Let N project onto the unavailable resource state, so that $I - N$ represents availability, and write the occupation bias relative to half occupation as $\xi = \langle N \rangle - \frac{1}{2}$. In the established interaction window with displacement paused and the preparation specified above, execute the entire sequence from source–auxiliary readout to resource swap with input-independent probability ε. The unexecuted branch bypasses the readout and retains the old resource. The bias changes by

$$\boxed{\Delta\xi = -\varepsilon\xi + \frac{\varepsilon}{2}\langle \sin(2\theta F_f) \rangle .}$$

(3.7)

Here θ is the accumulated interaction parameter, and the expectation is evaluated in the actual input. Resource replacement removes the old bias in the first term, while the matter response writes a new bias in the second. The displaced resource and auxiliary outputs remain in the joint state, together with their correlations. The background change therefore arises from the actual action of matter and affects its subsequent maintenance.

As matter moves, later maintenance may encounter these altered resources again or encounter others. We must therefore relate background change to actual contact. Take a working pool of N_{pool} resources. Let N_y be the occupation projection of resource y, and W_y the projection onto positions where matter contacts it. Each position contacts M resources, so $\sum_y W_y = MI$. Position and resource projections act on different factors. Averaging over the resources currently in contact gives a concrete realization of the availability operator B in Section 3.2:

$$B_{\text{cont}} = \frac{1}{M}\sum_y W_y(I - N_y).$$

(3.8)

Mean occupation of the entire pool describes its overall state, whereas actual maintenance depends on the part contacted at that moment. To relate them, let X_{av} be the mean bias of the pool and C_{cont} the difference between the contact average and the pool average:

$$
\begin{aligned}
X_{\mathrm{av}} &= \frac{1}{N_{\mathrm{pool}}} \sum_y \left(N_y - \tfrac{1}{2} I \right), \\
C_{\mathrm{cont}} &= \frac{1}{M} \sum_y W_y \left(N_y - \tfrac{1}{2} I \right) - X_{\mathrm{av}}.
\end{aligned}
$$

(3.9)

Substituting both terms into the availability gives

$$
\boxed{B_{\mathrm{cont}} = \frac{1}{2} I - X_{\mathrm{av}} - C_{\mathrm{cont}}.}
$$

(3.10)

This decomposition separates overall occupation from contact differences. Motion can bring matter back to its old background or toward fresh resources, changing its maintenance action even if the pool average remains fixed.

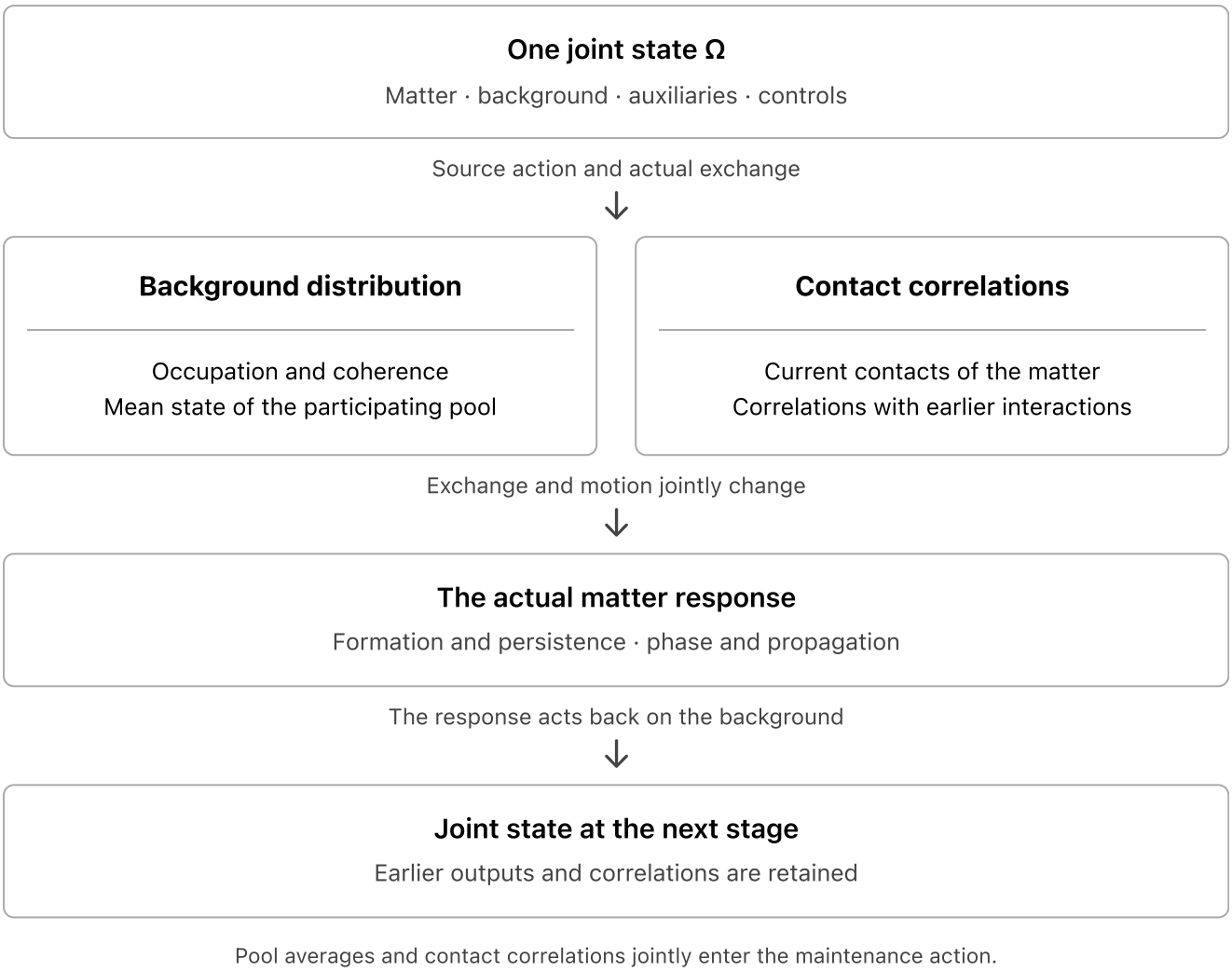


Figure 3-2. Feedback between matter and background. Matter alters the background; transport, exchange, and renewed contact then change the conditions of maintenance. Each stage takes the actual output of the preceding stage, retaining correlations among source, background, and auxiliaries in the next change.

Source response, resource transfer, and renewed contact affect one another, changing subsequent transfer probabilities and phase responses from cycle to cycle.

3.4 Dynamics of the Whole

Exchange, persistent evolution, and background feedback all belong to the evolution of the same joint state. Connecting the actual stages gives the complete joint process established in state-space dynamics. [1] Consider a cycle containing free evolution, exchange, return, and restoration. Free evolution advances phases, exchange transfers components, return retrieves temporarily stored matter components, and restoration uses prepared auxiliaries to establish a specified background preparation. Denoting the corresponding propagators by $U_{\mathrm{free}}, U_{\mathrm{ex}}, U_{\mathrm{ret}}, U_{\mathrm{rec}}$, we have

$$U_{\mathrm{cyc}} = U_{\mathrm{rec}} U_{\mathrm{ret}} U_{\mathrm{ex}} U_{\mathrm{free}}.$$

(3.11)

The factors act on the same joint space, with the identity on parts not involved in the current operation. Source writing, maintenance contacts, and protective control are included where they occur. Used sources, background, stores, and records remain in Ω_j. Let $P_{\mathrm{ready},\,j}$ denote fresh blank records initially uncorrelated with them. Successive cycles then give [1]

$$\Omega_{j+1} = U_{\mathrm{cyc}} \left(\Omega_j \otimes P_{\mathrm{ready},\,j} \right) U_{\mathrm{cyc}}^{\dagger}.$$

(3.12)

Transfer probabilities and phase responses in the next cycle therefore depend on the state left by the preceding one, while fresh preparations consumed by restoration are accounted for cycle by cycle.

Exchanges between matter carriers are incorporated in the same way. Section 3.1.2 of the unification theory implements a two-end operation between persistent carriers through temporary storage, coherent exchange, and inverse retrieval. [2] The two ends acquire new states and correlations while the mediator returns to its input state. The actual duration and control energy of these operations are included in the full process.

The full duration includes every stage that actually occurs. For serial stages of duration Δt_j,

$$T_{\mathrm{cyc}} = \sum_j \Delta t_j.$$

(3.13)

Simultaneous operations use the joint generator for the same interval, whose duration is counted once. Active evolution, return, and waiting each correspond to actual processes; the exchange angle of a single stage cannot replace the progress of the entire process.

Let $\mathbb{U}(t, t_0)$ be the propagator composed of all actual stages. With the convention $\hbar = 1$, the same dynamics takes the form

$$\begin{aligned} \Omega(t) &= \mathbb{U}(t, t_0) \Omega(t_0) \mathbb{U}^{\dagger}(t, t_0), \\ i\dot{\Omega}(t) &= [H(t), \Omega(t)]. \end{aligned}$$

(3.14)

Here t is the evolution parameter of the complete process; Chapter 4 establishes its relation to internal matter references through time measurement. The Hamiltonian $H(t)$ includes the free terms, maintenance, motion, exchange, and control of the actual participants. The staged realization and the continuous representation organize the same process in different ways.

The composition of matter in the universe changes within these processes as well. If P_f projects onto the matter carrier being tracked, its population $q_f = \mathrm{Tr}(\Omega P_f)$ satisfies

$$\dot{q}_f = i\mathrm{Tr}(\Omega[H, P_f]).$$

(3.15)

Maintenance terms commuting with P_f preserve its carrier population, while exchanges connecting to other modes or the background change it. Formation, persistence, and conversion are expressed by the same joint generator; feedback determines their succession as the input changes.

The evolution of the whole also determines changes of physical quantities in each observational channel. Using the common representation π of Chapter 2,

$$\omega_t(A) = \mathrm{Tr}[\Omega(t)\pi(A)], \qquad \omega_{p,t} = \omega_t|_{\mathfrak{A}_p}.$$

(3.16)

On differentiable stages for which the corresponding commutator expectation exists,

$$\boxed{\begin{aligned} &\tfrac{d}{dt}\omega_{p,t}(A) = i\mathrm{Tr}\big(\Omega(t)[H(t), \pi(A)]\big), \\ &A \in \mathfrak{A}_p. \end{aligned}}$$

(3.17)

The right-hand side generally contains joint expectations across components. Changes in the channels are therefore connected by the common state and its dynamics; a single-channel marginal need not obey a closed evolution equation. The unity of the universe is expressed here by an explicit dynamical relation: matter populations, background properties, and changes in all channels are obtained from one continuously evolving joint state. Energy transfers between them must obey the same balance.

3.5 Energy Balance and Thermal Evolution

Exchange redistributes energy between matter and background. When one part loses energy, the other participants receiving that change must be tracked as well. The parent spectrum provides a common energy basis, finite shell energies specify local exchange gaps, and the persistent generator of matter gives its actual phase spectrum. A concrete realization and calibration connect these three descriptions, while interaction and control energy enter the same balance. [1,2]

Let the joint Hamiltonian be $H = \sum_A H_A + V$, where H_A is the bare energy of each sector with no explicit time dependence. The joint dynamics gives the sector energy currents and, for a time-independent total Hamiltonian, the balance

$$J_A = \frac{d}{dt}\langle H_A\rangle = i\mathrm{Tr}(\Omega[H, H_A]),$$
$$\boxed{\sum_A J_A + \frac{d}{dt}\langle V\rangle = 0.}$$

(3.18)

Energy leaving one sector is received through changes in other sectors and in interaction energy. Staged control also supplies switching work: at a switch where the state is continuous, $W_j = \mathrm{Tr}[\Omega_j(H_{\mathrm{after}} - H_{\mathrm{before}})]$. This contribution is included as in Eq. (4.39) of the unification theory; when the controller is included in an autonomous description, it becomes energy transfer between participants. [2] Complete passivity of equilibrium states and constraints on state conversion under energy-conserving operations provide a reference for distinguishing thermal energy from usable work resources. [8,9]

Connecting heat exchange to matter maintenance requires balancing both energy changes. Let the two levels of the thermal receiver A be separated by $E_*\log 2$, while the two logical states of resource C have equal bare energy and the matter maintenance term is $E_* S(I - N_C)$. Write $S = \sum_\lambda \lambda P_\lambda$, retaining its positive and negative spectral branches. When the resource changes from unavailable to available, the maintenance energy in that branch increases by $E_*\lambda$; the receiver's bare thermal gap does not account for the entire change.

A finite realization uses a thermal-energy compensation store W_T and a maintenance-energy compensation store W_S. Choose equal-charge auxiliary ports and a prepared store support on which all required translations are accessible. Define the partial isometry L by

$$\begin{aligned}&|\lambda; 1_A, 1_C; w_T, w_S\rangle\\ &\quad\mapsto |\lambda; 2_A, 0_C; w_T - E_*\log 2, w_S - E_*\lambda\rangle.\end{aligned}$$

(3.19)

Here w_T, w_S are actual store energy levels. The reverse branch is given by $L^\dagger$, and unconnected states bypass the operation. The bare thermal change at the endpoints is compensated by W_T, and the maintenance-energy change by W_S; on a negative matter branch, the latter store gains energy. This forward restoration is powered by the prepared stores, while net energy removal from the thermal sector occurs in the preceding cooling stage. The receiver's actual occupation also determines the direction of later service. With H_{bare} including both stores and all other bare energies, the service window with displacement paused satisfies

$$\begin{aligned}H_D &= H_{\mathrm{bare}} + E_* S(I - N_C), \qquad [H_D, L] = 0,\\ H_{\mathrm{serv}} &= H_D + \hbar\Omega(L + L^\dagger), \qquad T_{\mathrm{serv}} = \frac{\pi}{2\Omega}.\end{aligned}$$

(3.20)

On each paired branch, this actual propagation completes the transfer while retaining the free phase. It acts branch by branch on the original joint state, carrying coherence between positive and negative matter branches into the successor together with conditional store translations. Preparation of finite stores, switching before and after service, and saved records are all included in the full process. If other noncommuting motion operations are enabled, their control work must also be included. Heat reception and resource restoration can thus be connected; the direction and amount of net restoration in each cycle remain determined by the actual input.

For a component described by a thermal state, temperature summarizes its energy distribution over states. Its role in exchange follows directly from the endpoint transfer in Section 3.1. For an initially uncorrelated preparation diagonal at the two endpoints, let the system populations be p_{ν}, p_{a} and the environment populations t_{u}, t_{v}. Then

$$\Delta p_{1} = \sin^{2}\alpha\,(p_{a}t_{v} - p_{\nu}t_{u}).$$

(3.21)

If the environment is in thermal equilibrium at physical inverse temperature $\beta_{\mathrm{phys}} = 1/(k_{\mathrm{B}}T)$, then $t_{u}/t_{v} = e^{-\beta_{\mathrm{phys}}(E_{a} - E_{\nu})}$. Exchange equilibrium accordingly requires $p_{a}/p_{\nu} = e^{-\beta_{\mathrm{phys}}(E_{a} - E_{\nu})}$. Ports with different energy gaps can share the same inverse temperature, although their actual approach to equilibrium is governed by their respective exchanges and common correlations.

For the thermal sector B, use the parent thermal state τ_{β} and energy unit E_{*} of Section 1.3, with $\beta = E_{*}\beta_{\mathrm{phys}} > 1$. Its mean energy is

$$\begin{aligned} U(\beta) &= \mathrm{Tr}(H_{\log}\tau_{\beta}) = -\frac{d}{d\beta}\log\zeta(\beta), \\ \langle E\rangle_{\beta} &= E_{*}U(\beta). \end{aligned}$$

(3.22)

Changing the thermal parameter changes the weights of different energy components. Differentiating the normalized thermal state and taking the energy expectation gives

$$\begin{aligned} \partial_{\beta}\tau_{\beta} &= (U - H_{\log})\tau_{\beta}, \\ \frac{dU}{d\beta} &= -C_{\beta}, \qquad C_{\beta} := \mathrm{Var}_{\tau_{\beta}}(H_{\log}). \end{aligned}$$

(3.23)

If the thermal sector follows $\rho_{B}(t) = \tau_{\beta(t)}$ during the stage considered, its outgoing power normalized by E_{*}, $J = -\dot{U}$, satisfies

$$\boxed{J = C_\beta \dot{\beta}.}$$

(3.24)

This relation connects temperature change to actual exchange: the energy current determines the rate of progress, while energy fluctuations determine how much temperature change the same current produces. For $J > 0$, β increases and temperature decreases. Thermal stages can therefore be ordered by increasing inverse temperature, with their durations determined by the actual dynamics.

Outputs departing from the Gibbs distribution continue to be described by the joint state Ω; temperature is used within a controlled thermal-state approximation.

Converting this thermal trajectory to physical temperature requires a common energy unit. The unification theory fixes the common response normalization through the sum of squared logarithmic intervals in both directions over the full background: [2]

$$\begin{aligned} C^* &= 2\sum_{n=1}^{\infty} \log^2\left(1+\frac{1}{n}\right) \\ &= 1.954378366537873\ldots, \\ E(W) &= E_0\sqrt{\frac{W}{C^*}}\log W. \end{aligned}$$

(3.25)

Here $W > 1$ is the participation weight of the specified response, and E_0 is the common energy calibration. Using the layer weights for $p = 7$ and depth $K = 41$ from Chapter 6 of that theory gives

$$\begin{aligned} \mathcal{L}_7 &= \frac{3}{2|\zeta(1/2)|} = 1.027147854134905\ldots, \\ W_{\mathrm{met}} &= 7^{41}\mathcal{L}_7 \\ &= 4.577755612508\ldots \times 10^{34}. \end{aligned}$$

(3.26)

The factor $\mathcal{L}_7$ corrects the absolute participation weight according to the prescribed layer structure, leaving the normalized uniform state unchanged. Taking $E_0 = 1\,\mathrm{GeV}$ and calibrating the parent spectrum by this metric energy scale yields

$$\boxed{\begin{aligned} E_* &= E_0\sqrt{\frac{W_{\mathrm{met}}}{C^*}}\log W_{\mathrm{met}} \\ &= 1.221446368809\ldots \times 10^{19}\,\mathrm{GeV}. \end{aligned}}$$

(3.27)

These constants connect the dimensionless parameter of the parent spectrum to physical temperature; their values inherit the same layer weighting, depth, and energy calibration. If one additionally adopts the prescription of one resolution unit per unit participation weight, its inverse weight is $\Delta\beta_{\mathrm{res}} = W_{\mathrm{met}}^{-1} \simeq 2.18448 \times 10^{-35}$. This specifies a resolution step in the thermal parameter; the time taken to traverse that interval is still determined by energy exchange.

An explicit class of collective exchanges further determines a cooling rate. Take $N \geq 2$ thermal participants, initially prepared in a product of truncated thermal states at the same temperature, each retaining energy levels labeled by integers $1 \leq m_i \leq M$. They exchange energy within the same whole. Let s measure exchange progress, and let the descent rate of participant i from m_i to m_i/d be controlled by the actual energies of the others:

$$a_{i,d}(\boldsymbol{m}) = \frac{\kappa\Lambda(d)}{N-1} \sum_{j \neq i} \log m_j,$$
$$d \mid m_i, \quad d > 1, \quad \kappa > 0.$$

(3.28)

Here q is prime, $r \geq 1$, $\Lambda(q^r) = \log q$, and Λ vanishes at all other integers. An actual joint gate implements this descent rate. Write the total rate as $r(\boldsymbol{m}) = \sum_{i,d} a_{i,d}(\boldsymbol{m})$, and let $|1\rangle$ and $|0\rangle$ denote the prepared states of the working receiver and the current record, respectively. For a progress step $\delta > 0$,

$$\begin{aligned} U_\delta|\boldsymbol{m}, 1, 0\rangle = \sqrt{1 - \delta r(\boldsymbol{m})}\ |\boldsymbol{m}, 1, 0\rangle \\ + \sum_{i \neq j} \sum_{\substack{d \mid m_i \\ d > 1}} \sqrt{\frac{\delta\kappa\Lambda(d)\log m_j}{N-1}} \\ \times |\boldsymbol{m}^{i \to m_i/d}, d, (i, j, d)\rangle. \end{aligned}$$

(3.29)

Each descent transfers $E_*\log d$ into the working receiver and leaves a label $\alpha = (i, j, d)$. The output and this label uniquely recover the input, so images of distinct inputs are orthogonal. Since $r(\boldsymbol{m}) \leq \kappa N(\log M)^2$, the condition $\delta\kappa N(\log M)^2 \leq 1$ ensures normalization. On a block with $r(\boldsymbol{m}) > 0$, rotate between the prepared vector and the normalized jump vector of equal total energy; use the identity on a zero-rate block. This gives an energy-conserving unitary extension. The same label then controls the energy-conserving swap $|d, a, \alpha\rangle \leftrightarrow |1, ad, \alpha\rangle$, whose entries denote receiver, energy store, and record, with a the integer label of the store level. The store starts at $a = 1$ and accommodates $1 \leq a \leq M^N$, sufficient for all initial thermal energy of this finite system. The receiver thereby returns to its prepared state, energy is transferred to the store, and correlations of the thermal participants with the store and old records remain. For the energy-diagonal thermal input considered here, these gates generate the stated population descent rates step by step; general coherent inputs still propagate under the full joint unitary.

This descent structure is compatible with the parent thermal distribution because

$$\sum_{d \mid m} \Lambda(d) = \log m, \qquad \sum_{d \geq 2} \frac{\Lambda(d)}{d^{\beta}} = U(\beta).$$

(3.30)

The first identity gives the total strength leaving level m, and the second the thermally averaged strength entering it from higher levels. Thus, on the untruncated parent spectrum, the unit descent generator $\mathcal{L}^{*}$ satisfies, for $p_{\beta}(m) = m^{-\beta}/\zeta(\beta)$,

$$(\mathcal{L}^{*} p_{\beta})(m) = [U(\beta) - \log m] p_{\beta}(m) = \partial_{\beta} p_{\beta}(m).$$

(3.31)

The mean energy of the other participants tends to $U(\beta)$ in the collective limit, giving the comparison trajectory $d\beta_{c}/ds = \kappa U(\beta_{c})$ in exchange progress s. Finite-participant outputs remain correlated. Once $\beta_{0} > 1$ and a finite progress interval are fixed, the local thermal-state errors from the participant number and spectral cutoff can be controlled separately. The actual exchange approximately maintains a common temperature without repreparing thermal states at every cycle.

The execution process determines the relation between exchange progress and elapsed time. Using a step δ satisfying the normalization condition above, place exchange and transfer to the store at successive program positions. The program clock has nearest-neighbor coupling γ_{clk}. Its input is a directed cosine packet of integer width $w \geq 2$ with carrier phase $e^{-i\pi j/2}$. Relative to the packet center, its envelope is $[1 + \cos(\pi j/w)]/\sqrt{3w}$ for $|j| \leq w$ and zero outside; the entire initial packet lies in an identity preparation segment before the execution region. During an execution interval with controlled endpoint effects, its mean forward speed is

$$v_{\mathrm{clk}} = \gamma_{\mathrm{clk}}(2 - d_{w}), \qquad d_{w} = \frac{2[1 - \cos(\pi/w)]}{3}.$$

(3.32)

A complete cycle contains two operations, exchange and transfer to the store, so comparison progress advances at speed $\delta v_{\mathrm{clk}}/2$. Substituting this into the temperature equation gives

$$\boxed{\begin{aligned} \frac{d\beta_{c}}{dt} &= \frac{U(\beta_{c})}{t_{*}}, \\ t_{*} &= \frac{2}{\kappa \delta v_{\mathrm{clk}}} = \frac{1}{\kappa \delta \gamma_{\mathrm{clk}}(1 - d_{w}/2)}. \end{aligned}}$$

(3.33)

The time scale t_{*} is determined jointly by cross-control, a single exchange, and the program clock. Let t_{0} be the time at which the mean progress enters the execution region. After $\beta_{c}(t_{0}) = \beta_{0}$,

$$\boxed{t - t_0 = t_* \int_{\beta_0}^{\beta_c(t)} \frac{db}{U(b)}.}$$

(3.34)

The actual clock has finite spreading, and the thermal sector is the marginal of joint outputs at different stages of progress. This time law describes the thermal trajectory it follows over a finite window. Local error is controlled by the finite participant number, parent-spectrum tail, discrete exchanges, clock spreading, and endpoint effects. Initial blank records, storage capacity, and time spent in the preparation region all belong to this realization.

The instantaneous energy current can also be obtained directly from the same program. Let E_r be the total thermal-sector energy after r exchanges, $q_r = E_r - E_{r+1}$, and $\psi_j(t)$ the amplitude at program position j. Only exchange edges change the thermal-sector energy, so

$$\begin{aligned} I_j(t) &= -2\gamma_{\mathrm{clk}} \mathrm{Im}[\bar{\psi_j}(t)\psi_{j+1}(t)], \\ \mathcal{P}_{\mathrm{out}}(t) &= \sum_{r=0}^{n-1} I_{2r}(t) q_r. \end{aligned}$$

(3.35)

Here n is the number of complete cycles in the program. The expression retains local returning probability currents, with energy received by the receiver and store. Normalizing total power by NE_*, the quantity $\mathcal{P}_{\mathrm{out}}/(NE_*)$ has the same dimensions as $C_{\beta_c} U(\beta_c)/t_*$ on the reference trajectory; the difference between their interval averages is controlled by endpoint energy errors. Both the cooling time coefficient and the heat balance thus have an explicit process basis, with fluctuations of the finite realization retained in the same description.

When the output need not remain thermal, heat can still be calculated directly from the complete cycle. Consider another preparation already containing matter, thermal participants, resources, and finite stores. At entry, the receiver has $A = 1$, the working record is ready, and the stores lie in the specified canonical support. Thermal exchange, cooling, archiving, source readout, resource swap, occupation recording, restoration, and further archiving occur in succession; the final stage preserves the entry support required for the next cycle. Fresh blank records are prepared each cycle. Archived records never accessed again are retained, while old records still used as controls, execution flags, and all used sources remain in the core, whose marginal satisfies $\rho_{j+1} = \Phi(\rho_j)$. This reduction traces out only records on which no later operation acts; Φ still fully depends on correlations among matter, background, and used sources within the core.

The cycle tracks the bare thermal energy of participant T and the selected shell endpoints. In units of E_*, write $U_{\mathrm{th}} = H_{\log,T} + (\log 2) P_e$, where $H_{\log,T}|m\rangle = (\log m)|m\rangle$ and P_e projects onto the excited endpoint with gap $E_* \log 2$. Receiver energy, store energy, and matter maintenance energy are counted separately. Thermal contact preserves this sum, and only cooling transfers it out. The complete cycle therefore gives

$$\boxed{\begin{aligned}\Phi^{\dagger}(U_{\mathrm{th}}) &= U_{\mathrm{th}} - D, \qquad D \geq 0,\\ \langle U_{\mathrm{th}}\rangle_n &= \langle U_{\mathrm{th}}\rangle_0 - \textstyle\sum_{j=0}^{n-1} \mathrm{Tr}(D\rho_j).\end{aligned}}$$

(3.36)

On the ready input space above, $D = U_{\mathrm{hot}}^{\dagger} Q_c U_{\mathrm{hot}}$, where $Q_c \geq 0$ is the heat-loss operator of the cooling stage and U_{hot} is the preceding thermal exchange; other stages preserve this bare thermal energy. No Gibbs state needs to be reprepared each cycle. Source pointer B and resource C obey a separate occupation balance. If source readout increases the B occupation by I_j^{src}, the swap preserves $N_B + N_C$, and restoration from the $A = 1$ input reduces the C occupation by $R_j \geq 0$, then

$$\Delta_j \langle N_B + N_C\rangle = I_j^{\mathrm{src}} - R_j, \qquad \sum_{j<n} R_j = \sum_{j<n} I_j^{\mathrm{src}} + \langle N_B + N_C\rangle_0 - \langle N_B + N_C\rangle_n.$$

(3.37)

A finite occupation stock permits finite net restoration; sustained mean restoration requires continued source input, with I_j^{src} updated from the current correlated state each cycle. Occupation, bare thermal energy, and maintenance-energy balances jointly describe this nonthermal loaded cycle. It and the collective thermal trajectory above are two explicit realizations. Receiver capacity and the used receiver state also enter the exact entropy balance of a Landauer process. [10]

As energy is received by other participants, their states and mutual relations change. An entropy balance expresses this accompanying change. For a general fixed thermal-sector energy H_B, take its initial marginal to be the Gibbs state $\tau_{\beta_{\mathrm{phys}}}$, and denote the remaining participants by A. Assume joint unitary evolution, set $Q = -\Delta E_B$, and measure entropy with $k_{\mathrm{B}} = 1$. For an initially uncorrelated finite environment, entropy production decomposes into correlation buildup and departure from the thermal state. [11] Using the joint-entropy and relative-entropy relations in Chapter 8 of state-space dynamics, and retaining initial correlations, we obtain [1]

$$\boxed{\begin{aligned}\Delta S_A = \beta_{\mathrm{phys}} Q + D(\rho_B' \| \tau_{\beta_{\mathrm{phys}}})\\ +\Delta I(A\!:\!B).\end{aligned}}$$

(3.38)

Here D is quantum relative entropy, measuring the final state's departure from the initial thermal state, and I is mutual information, describing correlations between the two parts. The final state may be out of thermal equilibrium. For initially uncorrelated inputs, $\Delta I = I_{\mathrm{out}} \geq 0$. The receiver accepts energy and also accommodates the loss of thermal-sector entropy and changes in correlation structure. Used auxiliaries have new states; their capacity to receive further energy depends on the next actual operation.

The universe evolves through continuous change of its common state. Thermal evolution describes the part that admits a temperature characterization. Matter formation, background change, and energy transfer also leave differences that can be preserved. Differences distinguishable by subsequent operations and persistently retained constitute records of this evolution.

Chapter 4. Records of the Universe

Joint cosmic evolution changes matter, background, and their relations together. Some changes occur only during the current interaction; others leave traces that later operations can distinguish. Preserved traces constitute records. When these records are actually read, events concerning the universe's own changes arise within it.

State-space dynamics places records, events, and observational spacetime within the same dynamics. [1] The unification theory further supplies the physical carriers of phase references, position motion, and light signals. [2] Records become events through readout, with each channel obtaining its observational content from the same whole. Matter clocks and light signals then establish temporal and spatial relations between events.

4.1 Formation and Preservation of Records

A record requires a change to remain distinguishable in subsequent processes. The storage and preservation of readable information in the environment are central issues in decoherence and quantum Darwinism. [12] Here we construct records from specific joint interactions and the intervals over which their outputs are preserved. Consider an interaction between a system S and storage degrees of freedom R. On a set of orthogonal states distinguishable by the interaction, let

$$U_{SR}(|s_\alpha\rangle \otimes |r_0\rangle) = |s'_\alpha\rangle \otimes |r_\alpha\rangle.$$

(4.1)

Linearity maps a coherent input to

$$\sum_\alpha c_\alpha |s_\alpha\rangle|r_0\rangle \mapsto |\Psi_{SR}\rangle = \sum_\alpha c_\alpha |s'_\alpha\rangle|r_\alpha\rangle.$$

(4.2)

Different branches leave different states in R. If their supports are orthogonal, a family of record projections $\{R_\alpha\}$ distinguishes them:

$$R_\alpha |r_\beta\rangle = \delta_{\alpha\beta}|r_\beta\rangle, \qquad \sum_\alpha R_\alpha = I_R.$$

(4.3)

Preservation requires subsequent operations to retain this distinction over the interval used. For example, if a storage propagator V preserves each record subspace, $V^\dagger R_\alpha V = R_\alpha$, the record yields the same result when read later. Actual storage may have a finite lifetime, with its usable range determined

by subspace mixing and information loss.

Retaining a record also changes the reduced coherence of the system itself. Taking the partial trace of the joint pure state above gives

$$\rho_S = \sum_{\alpha,\beta} c_\alpha c_\beta^* \langle r_\beta | r_\alpha \rangle |s'_\alpha\rangle\langle s'_\beta|.$$

(4.4)

Overlaps of record states determine the weights of the corresponding cross terms in the reduced system state. Orthogonal records eliminate these cross terms, while the complete joint state $|\Psi_{SR}\rangle$ retains phase relations between branches. Reading the system alone may therefore miss information preserved in the full state.

Records may reside in return auxiliaries, the thermal environment, propagation carriers, or dedicated instruments. Actual interactions leave them as distinguishable output differences that can be preserved. Matter in the universe thus participates in evolution and can also carry information about other processes. Coupling that information to an actual instrument makes it a readable result.

4.2 Events and Observational Spacetime

A preserved record becomes an event through actual readout. Let Ω_{out} be the joint state after the instrument interaction but before selection of an outcome, and let R_r be the record projection for outcome r. The event probability and unnormalized conditional state are [1]

$$\begin{aligned} p(r) &= \mathrm{Tr}(R_r \Omega_{\text{out}}), \\ \tilde{\Omega}_r &= R_r \Omega_{\text{out}} R_r, \qquad \sum_r R_r = I. \end{aligned}$$

(4.5)

For nonzero probability, the subsequent conditional state is $\Omega_r = \tilde{\Omega}_r / p(r)$. The full outcome set includes no response, exit, and other distinguishable outputs. Readout back-action, correlations between participants, and auxiliaries that will act again are retained in the conditional state.

Joint propagation, pointer projection, and the conditional successor define a completely positive quantum instrument; its dilation and measurement-process formulations are described in Refs. [13,14,15]. Successive observations inherit these conditional states. If an outcome changes later connections or durations, those changes enter the next actual stage. Chapter 6 of state-space dynamics gives the composition of quantum instruments and event sequences, keeping each result connected to its subsequent operations. [1]

Readout gives outcomes for physical quantities, while the corresponding observable algebra expresses which outcomes can be distinguished and how they are related. Records and events thereby connect to the structure of observational spacetime. Chapter 5 of state-space dynamics establishes a basic unit from position resolution and negation: [1] for a nonzero residue class a at an odd prime p,

$$D_a = \{a, -a\}, \qquad \mathcal{H}_{D_a} = \mathrm{span}\{|a\rangle, |-a\rangle\}.$$

(4.6)

Position projections distinguish the two branches and negation connects them, jointly generating $M_2(\mathbb{C})$. Its self-adjoint part is $\mathrm{Herm}_2(\mathbb{C})$, where two diagonal entries and one complex off-diagonal entry give four independent real parameters. Every self-adjoint observable has the form

$$A = uI_2 + \sum_{i=1}^{3} x_i \sigma_i, \qquad u, x_i \in \mathbb{R},$$

(4.7)

where σ_i are the Pauli matrices. Normalization at the identity and vanishing on rank-one positive operators determine the quadratic form established in Chapter 5 of state-space dynamics:

$$\boxed{q(A) = \det A = u^2 - |\boldsymbol{x}|^2.}$$

(4.8)

Its positive cone is $u \geq |\boldsymbol{x}|$, with geometric signature $(1, 3)$. Nondegenerate basic units in all odd-prime channels therefore have the same four-dimensional form. Real-linear bijections preserving positivity in both directions and preserving the identity retain this structure. The parameters p and K change what can be resolved, while the parent state space retains richer degrees of freedom. After compression of a state to a two-point unit, its trace gives that unit's population; matter references and propagation measurements further assign elapsed times and distances.

4.3 Relations between Observational Channels

The channels share a basic observational form, and their readings of the same object are related by the common state. This relation can first be calculated in the common finite space of Chapter 1. Take a finite set S containing at least two odd primes, let $M = \prod_{p \in S} p^{K_p}$, and choose a unit residue class a in the composite residue-class space. The global negation unit is spanned by $|a\rangle_M$ and $|-a\rangle_M$. Write $a_p = a \bmod p^{K_p}$, and choose $|+\rangle_p = |a_p\rangle$, $|-\rangle_p = |-a_p\rangle$ in each local unit. The Chinese remainder theorem represents the global unit as

$$\boxed{\begin{aligned} J|+\rangle &= \bigotimes_{p\in S} |+\rangle_p, \\ J|-\rangle &= \bigotimes_{p\in S} |-\rangle_p. \end{aligned}}$$

(4.9)

Here J is an isometric representation of the same finite unit in composite coordinates. The global positive branch determines all local positive branches simultaneously, and likewise for the negative branch. For a local projection $P_{p,\pm}$, extended by the identity on the remaining factors, direct calculation gives

$$P_{p,\pm} J = J P_{\pm}, \qquad J^{\dagger} P_{p,+} P_{q,-} J = 0 \quad (p \neq q).$$

(4.10)

Each end therefore reads the same global branch. The operator identity guarantees this correspondence for every input state of the unit.

Let $\rho = (\rho_{\epsilon\epsilon'})$ be a state on this two-dimensional unit, with joint representation $\Omega = J\rho J^{\dagger}$. Reading a single channel and tracing out the other factors removes the off-diagonal terms because a traced-out channel contributes $\langle -|+\rangle = 0$. Thus

$$\rho_p = \mathrm{Tr}_{S\setminus\{p\}} \Omega = \rho_{++} P_{p,+} + \rho_{--} P_{p,-}.$$

(4.11)

Local readings retain the populations of the common branches, while full coherence remains in the joint state. Let X_p, Y_p, Z_p be the Pauli actions on each local unit, and choose any $p_0 \in S$. Then

$$\begin{aligned} J^{\dagger} Z_{p_0} J &= \sigma_3, \\ J^{\dagger} \Big(\bigotimes_{p\in S} X_p \Big) J &= \sigma_1, \\ J^{\dagger} \Big(Y_{p_0} \otimes \bigotimes_{p\neq p_0} X_p \Big) J &= \sigma_2. \end{aligned}$$

(4.12)

Together with the identity, these three operators recover every self-adjoint observable of the common unit. The same four-dimensional observational content can thus be obtained jointly from local population readings and cross-channel coherence readings. The information of the whole established in Chapter 1 acquires an explicit geometric expression here.

The full representation preserves all observational content, whereas reduction to a single channel preserves its local statistics. The common finite unit displays this distinction directly: the Chinese remainder theorem determines local coordinate factors, and joint Pauli actions recover coherence across the ends. Native algebras containing scaling remain as described in Section 1.2.

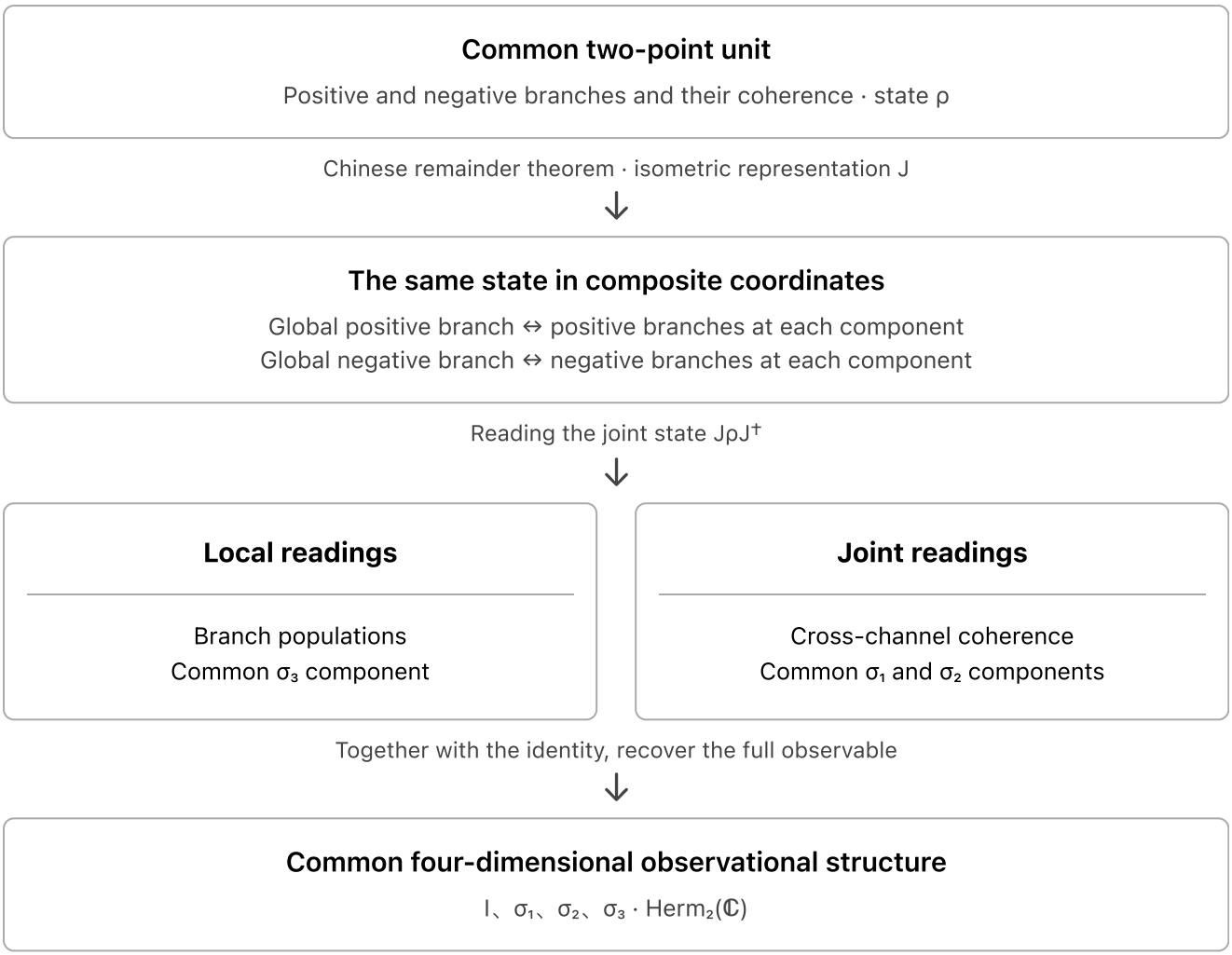


Figure 4-1. Cross-channel observation of a common unit. A global two-point unit has common positive and negative branches in composite coordinates. Local projections read branch populations, and joint Pauli actions read coherence; together they recover the same four-dimensional observable space. Arrows indicate representation and readout relations for the same unit.

The common unit above relates different readings of the same state. Correspondence between successive events additionally requires actual operations to preserve this relation stage by stage. For different channel realizations of the same process, let J_p be an isometry from the common logical carrier to the realization in channel p. Let $K_{p,r}$ and $K_{*,r}$ be the conditional operators with the actual fine records and participating auxiliaries retained. If the actual stages satisfy

$$K_{p,r}J_p = J_pK_{*,r}, \qquad J_p^\dagger J_p = I, \tag{4.13}$$

then composition along the same outcome sequence $\boldsymbol{r} = (r_1, \ldots, r_n)$ gives $K_{p,\boldsymbol{r}}J_p = J_pK_{*,\boldsymbol{r}}$. For a common input ρ,

$$\begin{aligned}\Pr_p(\boldsymbol{r} \mid \rho) &= \mathrm{Tr}\left[K_{p,\boldsymbol{r}} J_p \rho J_p^\dagger K_{p,\boldsymbol{r}}^\dagger\right] \\ &= \mathrm{Tr}\left[K_{*,\boldsymbol{r}} \rho K_{*,\boldsymbol{r}}^\dagger\right].\end{aligned}$$

(4.14)

The corresponding conditional states are also related by J_p, so the same event sequence retains its correspondence across realizations.

4.4 Matter Clocks and Time Calibration

The order of operations determines the order of events; actual matter references measure the time between them. Studies of quantum-clock limits in spacetime measurement and of autonomous timekeeping include clock dynamics, resources, and readout precision. [16,17] When two components of persistent matter retain coherence, their relative phase accumulates under maintenance. Channels capable of reading that coherence obtain the corresponding signals from the same process.

Section 2.4.6 of the unification theory relates matter phase to complete-cycle time measurement. [2] Suppose two components have distinct persistent spectral values E_a, E_b, an active window t_H, and other processing that pauses this phase for duration T_{proc}. Then

$$\tau_{\mathrm{cyc}} = t_H + T_{\mathrm{proc}}, \qquad f_{\mathrm{act}} = \frac{t_H}{\tau_{\mathrm{cyc}}}.$$

(4.15)

In the established scalar-survival realization, the selected coherence is multiplied each cycle by a positive real survival factor $0 < q_T \leq 1$. Let $c_{ab}^{(0)}$ be the nonzero initial coherence and $\tilde{\varphi}_{ab}$ its continuously unwrapped phase. After n complete cycles, the coherence and actual frequency are

$$c_{ab}^{(n)} = q_T^n e^{-in(E_a - E_b)t_H} c_{ab}^{(0)},$$

$$\boxed{\begin{aligned}\omega_{ab}^{\mathrm{obs}} &= -\frac{\Delta\tilde{\varphi}_{ab}}{n\tau_{\mathrm{cyc}}} \\ &= f_{\mathrm{act}}(E_a - E_b).\end{aligned}}$$

(4.16)

Cosine and sine readings of the phase are obtained under the same preparation, recovering the phase and its continuous change. The positive real survival factor attenuates amplitude without adding phase. Elapsed time is counted over the complete process of Section 3.4, with initial formation and loading counted whenever they occur. [1,2] On a readable interval of fixed frequency, $\Delta t = -\Delta\tilde{\varphi}/\omega^{\mathrm{obs}}$ gives the elapsed time.

Cosmic evolution also changes the reference itself. The two spectral components of a matter clock subject the background to different actions, so their remaining coherence must be retained together with the background. Tracking only the two matter populations no longer determines the subsequent phase.

Consider a protected maintenance window in which internal spectral labels are preserved. Let λ_a be a real eigenvalue of the current maintenance operator $M_f \otimes Y$, with spectral label $|a\rangle$. Collect all degrees of freedom other than this label into an environment E, with generator H_E, and use the actual contact availability B_{cont} from Chapter 3. Denote the environmental cross block between spectral components by R_{ab}; its trace z_{ab} is the coherence readable by the matter clock. In this window,

$$\begin{aligned} H_a &= H_E + \lambda_a B_{\mathrm{cont}}, \\ R_{ab} &= \langle a|\Omega|b\rangle, \qquad z_{ab} = \mathrm{Tr} R_{ab}. \end{aligned}$$

(4.17)

The cross block then satisfies $\dot{R}_{ab} = -i(H_a R_{ab} - R_{ab} H_b)$. Taking its trace cancels the common environment term, giving

$$\begin{aligned} \dot{z}_{ab} &= -i(\lambda_a - \lambda_b)\mathrm{Tr}(B_{\mathrm{cont}} R_{ab}), \\ \boxed{\omega_{ab} &:= -\frac{d}{dt}\arg z_{ab} \\ &= (\lambda_a - \lambda_b)\mathrm{Re}\frac{\mathrm{Tr}(B_{\mathrm{cont}} R_{ab})}{z_{ab}}.} \end{aligned}$$

(4.18)

For $\lambda_a \neq \lambda_b$, the phase is continuously unwrapped on a readable window with $z_{ab} \neq 0$; actual visibility and survival are obtained separately from the same output. The block R_{ab} retains matter back-action on the environment and environmental correlations, so it determines the phase rate. The contact decomposition of Chapter 3 further gives

$$\frac{\omega_{ab}}{\lambda_a - \lambda_b} = \frac{1}{2} - \mathrm{Re}\frac{\mathrm{Tr}[(X_{\mathrm{av}} + C_{\mathrm{cont}})R_{ab}]}{z_{ab}}.$$

(4.19)

Overall background occupation and actual contact relations jointly enter the matter phase. If other operations mix spectral labels, the coherence must be propagated with the complete generator; the mean resource occupation cannot in general replace this phase reading.

To compare different matter clocks, first fix their nonzero complete-cycle frequencies $\omega_{f,0}$ at entry. On the same complete readable interval $I = [t_1, t_2]$, define from the actual phase change

$$\omega_{f,I}^{\mathrm{read}} = -\frac{\tilde{\varphi}_f(t_2) - \tilde{\varphi}_f(t_1)}{|I|}, \qquad F_{f,I} = \frac{\omega_{f,I}^{\mathrm{read}}}{\omega_{f,0}}.$$

(4.20)

Paused stages still contribute to the elapsed time in the denominator. Any additional phase generated by those stages also enters the numerator. The input calibration thereby remains fixed in subsequent evolution, allowing actual comparison of background-induced changes.

Within a common resolution and visibility range, suppose the phases of several readable matter clocks are described by the same positive function $F(t)$, namely,

$$-\Delta\tilde{\varphi}_f(I) = \omega_{f,0} \int_I F(t)\, dt,$$

(4.21)

A common matter time then follows:

$$\boxed{d\tau = F(t)\, dt.}$$

(4.22)

Phase changes of different matter species thereby provide a common elapsed-time reference, in which light propagation and other processes can also be expressed. Deviations between clocks directly quantify the accuracy of common timekeeping.

Actual protection can realize this common timing. Let Q_{old} project onto a previous active record, and choose a lowering operator L_f for a pair in the actual motion spectrum such that $[H_{\mathrm{free}}, L_f] = -\Delta_f L_f$, using $\hbar = 1$. Read the coherence of the original matter through $z_f^{\mathrm{old}} = \mathrm{Tr}(\Omega Q_{\mathrm{old}} L_f)$. Each gate preserves the old record, and the nonfree parts of cooling and archiving commute with $Q_{\mathrm{old}} L_f$. Archiving after cooling routes that record branch into a subspace on which both the forward and reverse restoration couplings vanish, leaving free matter propagation unchanged. The complete cooling–archiving–restoration–archiving sequence therefore satisfies

$$z_f^{\mathrm{old}}(t) = e^{-i\Delta_f (t - t_0)} z_f^{\mathrm{old}}(t_0).$$

(4.23)

The two loaded matter clocks retain a common rate relative to their respective input calibrations while other branches extract heat and restore resources. The projection Q_{old} identifies the matter history being tracked; the reading remains a joint moment retaining all outcomes. Reconnecting source readout changes the original cross block, and in an example under the same load the two clocks can even change in opposite directions. Protection and source action thus give explicit processes in which common timing holds and in which it fails, respectively.

4.5 Light Signals and Spacetime Scales

Matter clocks compare elapsed times between local events, while light signals connect events at different positions.

Obtaining distant information includes actual emission and reception. Emitting matter couples phase or other distinguishable information to light degrees of freedom, propagation conveys it elsewhere, and receiving matter compares it with a local reference and leaves a record. Emitter, light, receiver, and instrument jointly evolve, with reception time and its distribution obtained from the actual operation. Experimental frequency shifts likewise compare local transition references at emitter and receiver, as in the gravitational frequency shift measured by Pound and Rebka. [18]

Use the massless photon channel identified in the unification theory. In the uniform shared propagation of Chapter 3, the position step ℓ retains the initial calibration of the earlier work. When maintenance and displacement are active together and paused together, the long-wavelength speed is [2]

$$c_{\mathrm{obs}} = f_{\mathrm{act}} g \ell .$$

(4.24)

This activity fraction uses the same complete cycle as the matter phase. Matter clocks and light propagation are connected by a common operation, while finite-wavenumber dispersion and actual directional connections are retained from the propagation process.

In a window where background change is negligible and propagation is uniform and nearly nondispersive, apparatus A emits a signal at local matter time τ_- and receives it at τ_+ after reflection from stationary apparatus B. Use the light speed c_0 calibrated in that window, include reflection and readout in the actual instrument, correct their delays, and assume identical outward and return propagation. Radar calibration is then

$$\boxed{\begin{aligned} R &= \tfrac{c_0}{2}(\tau_+ - \tau_-), \\ \tau_B^{\mathrm{rad}} &= \frac{\tau_+ + \tau_-}{2}. \end{aligned}}$$

(4.25)

Two records of the same apparatus determine the distance and assigned time of the distant event. Comparisons along different directions give the four-dimensional observational structure actual units of time and length.

As the background evolves, the propagation duration may also change. To compare such changes, keep the initial clock and position calibrations fixed. Set $F(t_0) = G(t_0) = 1$, and let χ be a fixed reference position coordinate. Where common matter time exists, propagation is uniform and isotropic, and

dispersion is negligible, let the positive function $G(t)$ describe the change in light propagation rate relative to entry. For a fixed reference position interval $\Delta\chi$, actual propagation and the emission and reception times satisfy

$$\frac{d\chi}{dt} = c_0 G(t),$$
$$\Delta\chi = c_0 \int_{t_e}^{t_r} G(t)\, dt.$$

(4.26)

Expressing the same light path in matter time $d\tau = F(t)\, dt$ gives the propagation rate

$$\frac{d\chi}{d\tau} = c_0 \frac{G(t)}{F(t)}.$$

(4.27)

Using common matter time and the initial light-speed unit, write the light path as $d\chi/d\tau = c_0/a$. The scale relative to entry is then

$$\boxed{a(t) = \frac{F(t)/G(t)}{F(t_0)/G(t_0)}.}$$

(4.28)

The function F comes from actual matter coherence and complete time accounting, and G from light propagation in the same process. If both are modulated only by a common activity factor, their ratio stays fixed; if the background affects them differently, the relative scale changes. Differences across directions or frequency bands determine the range in which a single scale description applies.

The same light path can therefore be written as $\Delta\chi = c_0 \int_{\tau_e}^{\tau_r} d\tau/a(\tau)$. Comparison of intervals between neighboring events provides another test of the relation between scale and actual signals. For individually identifiable emission–reception pairs at a fixed reference position interval, vary the emission time. If the mapping is differentiable, differentiating the endpoints of the light-path integral gives

$$\frac{dt_r}{dt_e} = \frac{G(t_e)}{G(t_r)}, \qquad \boxed{\frac{d\tau_r}{d\tau_e} = \frac{a(t_r)}{a(t_e)}.}$$

(4.29)

Intervals between neighboring events test the same scale relation, with propagation delay included through endpoint differentiation. In a further comparison of carrier frequencies, matter transitions and receiver response jointly determine the identifiable spectral lines.

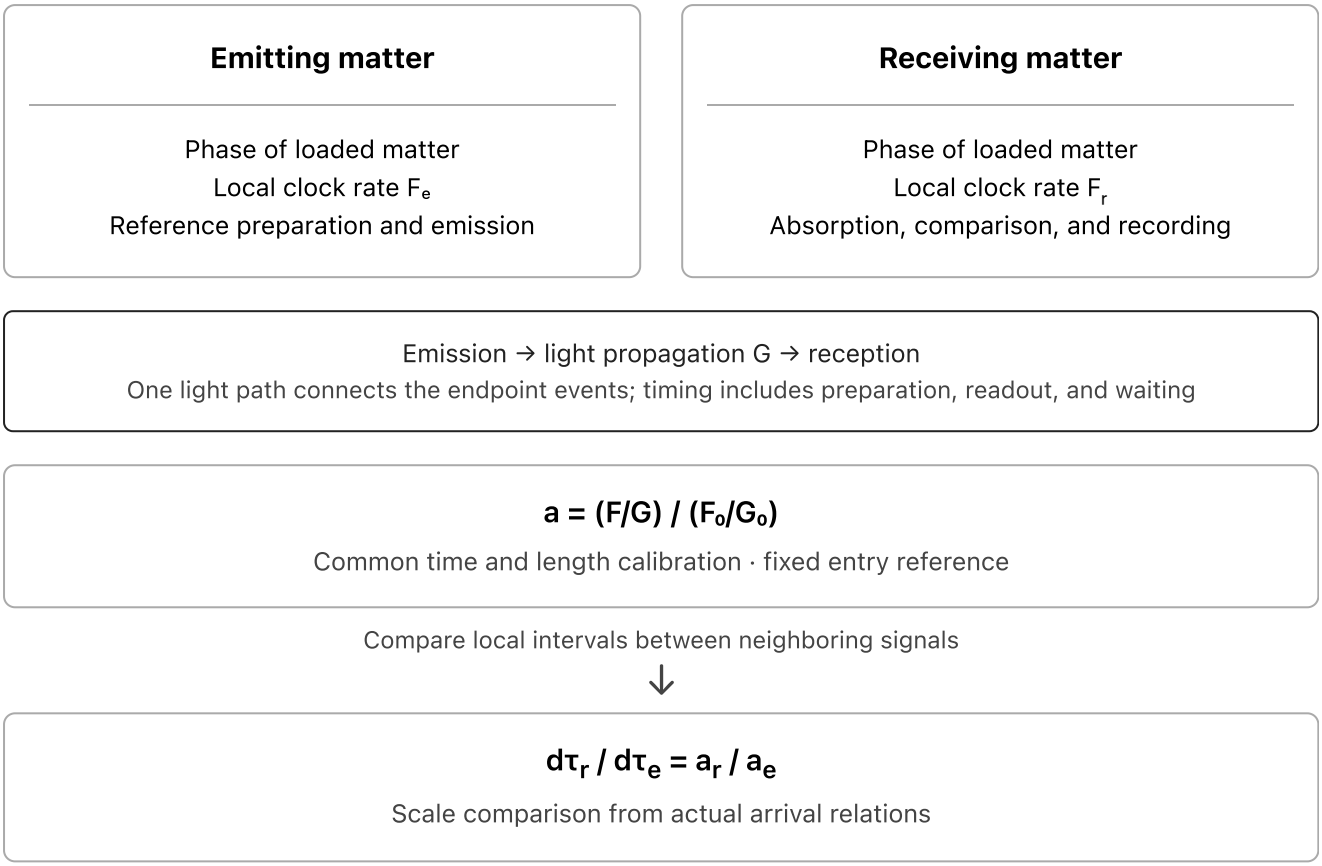


Figure 4-2. Emission, reception, and the common scale. Matter phases at both ends provide local time references, light propagation connects neighboring events, and the receiver records their comparison. Within a uniform, isotropic window of low dispersion, the arrival relation connects local time intervals to scale change.

Comparison of material spectral lines further involves coherence between different times in the emission window. Optical coherence and photon-counting theories relate fields to detection statistics through correlation functions. [19,20] The two-time kernel used here retains conditional correlations involving the actual source, resources, and reception process. Expand the source action to one emission insertion, and consider a participating source branch whose weight remains fixed during the emission window. Let $|\eta(s)\rangle$ be the normalized environment state following emission at s, transported to a common endpoint; mixed inputs can be treated together with their original purifiers. The two-time kernel

$$C(s', s) = \langle \eta(s') | \eta(s) \rangle \tag{4.30}$$

retains the overlap of source, resources, stores, and old records between two emission histories. Free light propagation preserves both mode frequencies and these conditional correlations. Suppose that, in a controlled comparison, the receiver has mode response $r_k(\delta)$, where δ is the instrument setting and κ_k the emission coupling weight. The click probability in this comparison is

$$w_\delta(s) = \sum_k r_k(\delta)\kappa_k e^{-i\omega_k(t_r - s)},$$
$$q_1 = p_{\mathrm{acc}} g_S^2 \int\int w_\delta(s) w_\delta(s')^* C(s', s)\, ds\, ds'.$$

(4.31)

Here p_{acc} is the weight participating in this effect within the original joint input, g_S is the emission strength, and the integrals cover the actual emission window. Modes are summed before taking the probability, so intermode interference and two-time background correlations both enter reception. The actual probability q is calculated from the full joint propagator and click projection. If the comparison error in the click vector is at most ε, the triangle inequality gives

$$\boxed{[\max(0, \sqrt{q_1} - \varepsilon)]^2 \le q \le (\sqrt{q_1} + \varepsilon)^2.}$$

(4.32)

Higher-order emission, the difference between the loaded receiver and the comparison operation, and readout error jointly enter this envelope, connecting spectral comparison to reception events. A terminal click describes the state at the end of reception; time records further resolve arrival durations.

Matter clocks and light signals give temporal and spatial relations to the records left by evolution. We now use these readings to examine explicit manifestations of thermal evolution, matter composition, and scale change.

Chapter 5. Cosmic Phenomena

The matter composition, spatial scale, and visible structures of the universe are different manifestations of the evolution of the same whole. Energy transfer in the background changes the conditions for matter formation and persistence, and the surviving matter continues to act on the background. Through matter clocks and light propagation, these processes yield comparable times, distances, and frequencies. Actual processes thus connect early and late states, while observations read the changes that can propagate and be preserved.

This chapter follows the early universe, matter composition, scale change, spatial distribution, and late evolution. During stages admitting a thermal description, increasing inverse temperature orders the cooling. The duration of each stage, the direction of scale change, and which matter survives are determined, respectively, by energy currents, phases, and survival within the same joint dynamics.

5.1 The Big Bang and the Early Universe

The temperature and anisotropies of the cosmic microwave background constrain early thermal evolution observationally. [21,22] Within the present framework, the early universe first concerns the high-temperature structure of the common background. Using the energy unit E_* of Chapter 3, $\beta = E_*/(k_{\mathrm{B}}T)$. The Bost–Connes system has a unique KMS state for $0 < \beta \leq 1$ and a family of extremal KMS states parameterized by $\widehat{\mathbb{Z}}^{\times}$ for $\beta > 1$. [1,3,4] The distinct states on the low-temperature side can be distinguished by corresponding observations, while still belonging to the same parent structure established in Chapter 1.

This phase transition provides a thermal-structural identification of the Big Bang: along cooling, the background passes through the critical region near $\beta = 1$, changing its phase structure and distinguishable responses. With the common energy calibration $E_* = 1.22144637\ldots \times 10^{19}$ GeV of Section 3.5, the corresponding critical temperature is $T_* = E_*/k_{\mathrm{B}}$. [2]

Energy participation near criticality follows directly from the parent spectrum. On the $\beta > 1$ side, level m has probability $p_m = m^{-\beta}/\zeta(\beta)$. Put $\varepsilon = \beta - 1$. Using $\zeta(1+\varepsilon) = \varepsilon^{-1} + \gamma + O(\varepsilon)$, the mean energy and thermal response of Chapter 3 give

$$\begin{aligned} U(1+\varepsilon) &= \frac{1}{\varepsilon} - \gamma + O(\varepsilon), \\ C_\beta &= \frac{1}{\varepsilon^2} + O(1). \end{aligned} \tag{5.1}$$

Here γ is the Euler–Mascheroni constant. As the critical point is approached, an increasingly broad spectrum contributes to the mean energy, so a small inverse-temperature change corresponds to substantial energy redistribution. The high-temperature side is described by the algebraic KMS

condition. [3,4,23]

The relation between actual finite participation and the parent-spectrum limit can also be made explicit. Truncate the integer spectrum at L, setting $Z_L = \sum_{m=1}^{L} m^{-\beta}$ and $U_L = -\partial_\beta \log Z_L$. At criticality, the harmonic sum and its logarithmically weighted counterparts give

$$\begin{aligned} Z_L(1) &= \log L + \gamma + O(L^{-1}), \\ \sum_{m=1}^{L} \frac{\log m}{m} &= \frac{1}{2}(\log L)^2 + O(1), \\ U_L(1) &= \frac{1}{2}\log L - \frac{\gamma}{2} + O\left(\frac{1}{\log L}\right). \end{aligned} \tag{5.2}$$

The response of every fixed participation range remains finite. Increasing the participating spectrum progressively reveals the critical enhancement of the parent spectrum. The integer cutoff illustrates this limit; shell structure and participation weights in a specific finite reduction are still calculated from its actual construction.

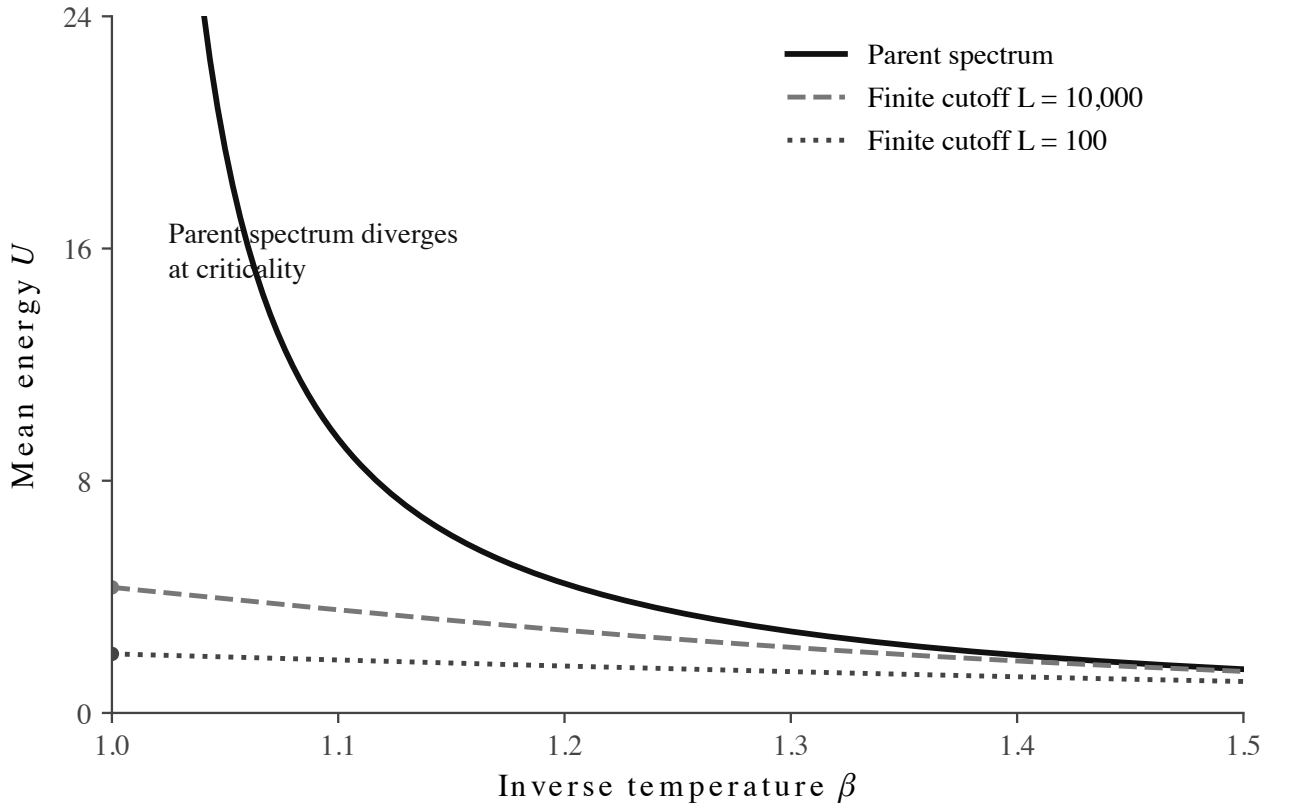


Figure 5-1. Energy participation in the critical region. The mean energy of the infinite parent spectrum diverges as $\beta \downarrow 1$, while two finite integer cutoffs remain finite at criticality. The horizontal axis orders thermal states, and the vertical axis gives mean energy normalized by E_*. The spectral cutoffs do not represent spatial boundaries of the universe.

The thermal structure must also be connected to internal elapsed time. On an interval of monotonic cooling where the thermal sector follows the same thermal-state family, Chapter 3 gives $J = C_\beta \dot{\beta}$ and Chapter 4 gives $d\tau = F dt$. Hence

$$\boxed{\tau(\beta_2) - \tau(\beta_1) = \int_{\beta_1}^{\beta_2} \frac{F(b) C_b}{J(b)}\, db, \qquad J > 0.}$$

(5.3)

Suppose the actual energy current and clock rate near criticality satisfy $J(1+\varepsilon) \sim A\varepsilon^{-r}$ and $F(1+\varepsilon) \sim B\varepsilon^{s}$, with $A, B > 0$. The integrand is then of order ε^{r+s-2}. The critical endpoint lies at finite matter time if $r + s > 1$; equality gives a logarithmic divergence. These exponents must come from the actual operations. The critical thermal structure alone does not determine whether the universe has a finite-time beginning, nor does it directly imply a singularity in spatial curvature or energy density.

For the collective-cooling comparison trajectory of Section 3.5, substituting the reference energy current $J_c = C_{\beta_c} U(\beta_c)/t_*$ gives the comparison duration

$$\Delta\tau_c = t_* \int_{\beta_1}^{\beta_2} \frac{F(b)}{U(b)}\, db.$$

(5.4)

where t_* comes from the actual exchange and program clock, and F from a matter reference in the same process. The critical form of this thermal integral can be calculated directly:

$$\begin{aligned} \Theta(\beta) &:= \int_1^\beta \frac{db}{U(b)} \\ &= \frac{\varepsilon^2}{2} + \frac{\gamma\varepsilon^3}{3} + O(\varepsilon^4), \\ \beta - 1 &= \sqrt{2\Theta} + O(\Theta), \\ \frac{T}{T_*} &= \frac{1}{1 + \sqrt{2\Theta} + O(\Theta)}. \end{aligned}$$

(5.5)

If this cooling relation continues toward the critical endpoint, the process parameter satisfies $t - t_c = t_*\Theta$, producing a square-root form of early cooling. If the matter clock rate also tends to a finite positive value F_c, then $\tau - \tau_c \sim F_c t_* \Theta$. The finite realization in Chapter 3 starts from fixed $\beta_0 > 1$; continuation to criticality still requires control of the expanding participating spectrum and the corresponding resources.

The near-critical background also contains directional differences between states. For an odd prime p, let $c \in (\mathbb{Z}/p\mathbb{Z})^\times$ be the modulo-p component of the extremal-state parameter. On the low-temperature side, the expectation of the phase observable $e(1/p)$ is [3,4]

$$\Phi_\beta(c) = \frac{1}{\zeta(\beta)} \sum_{m \geq 1} \frac{e^{2\pi i c m/p}}{m^\beta}.$$

(5.6)

Its real and imaginary parts are read separately. To compare directions, expand Φ_β on the unit group. For a nontrivial multiplicative character χ, the Gauss sum $\tau(\overline{\chi}) = \sum_c \chi(\bar{c}) e^{2\pi i c/p}$ satisfies

$$\sum_c \chi(\bar{c}) e^{2\pi i c m/p} = \chi(m)\tau(\overline{\chi}), \qquad |\tau(\overline{\chi})|^2 = p.$$

(5.7)

Here $\chi(m) = 0$ when $p \mid m$. Substituting into the expansion coefficient gives the directional component relative to the symmetric average:

$$\begin{aligned} \delta\Phi_\chi(c) &= A_\chi(\beta)\chi(c), \\ A_\chi(\beta) &= \frac{\tau(\overline{\chi})L(\beta,\chi)}{(p-1)\zeta(\beta)}. \end{aligned}$$

(5.8)

Under uniform averaging over c, character orthogonality makes the first moment vanish, while the second-order intensity is

$$\boxed{\begin{aligned} P_\chi(\beta) &:= \left\langle |\delta\Phi_\chi|^2 \right\rangle_c \\ &= \frac{p|L(\beta,\chi)|^2}{(p-1)^2\zeta(\beta)^2}. \end{aligned}}$$

(5.9)

The quantity P_χ measures directional differences between extremal states, whereas C_β measures energy fluctuations within the same thermal distribution. Uniform averaging here corresponds to a uniform preparation of extremal states; other preparations use their respective weights.

The L-function of a nontrivial character is analytic at $\beta = 1$. Define $B_\chi = \tau(\overline{\chi})L(1,\chi)/(p-1)$. Together with $1/\zeta(1+\varepsilon) = \varepsilon + O(\varepsilon^2)$, this gives

$$\boxed{\begin{aligned} \delta\Phi_\chi(c) &= B_\chi\chi(c)\varepsilon + O(\varepsilon^2), \\ P_\chi(1+\varepsilon) &= |B_\chi|^2\varepsilon^2 + O(\varepsilon^3). \end{aligned}}$$

(5.10)

Directional differences in the selected phase observable emerge continuously from criticality toward lower temperatures: the amplitude begins at first order, and the intensity at second order. As temperature decreases further along this thermal-state family, both $L(\beta, \chi)$ and $\zeta(\beta)$ tend to one, so

$$\begin{aligned} P_{\chi}(\infty) &= \frac{p}{(p-1)^2}, \\ P_{\chi}(\infty)\big|_{p=7} &= \frac{7}{36} = 0.194444\ldots. \end{aligned}$$

(5.11)

Finite-temperature values show how the directions approach this common limit. Take 3 as the primitive root modulo 7, with $\chi_j(3^a) = e^{2\pi i j a/6}$. Conjugate directions have equal second-order intensities, dividing the five nontrivial directions into three groups. Evaluation of the convergent L-function sums gives

β	χ_1, χ_5	χ_2, χ_4	χ_3
1.1	0.001413	0.000593	0.002451
2	0.062385	0.046647	0.095356
5	0.175971	0.174419	0.191000

Near criticality, directions have different growth coefficients; as temperature decreases, they approach the same low-temperature intensity. Equal limiting values therefore do not imply equality of all directions throughout evolution.

Higher-order directional correlations follow in the same way. For $p = 7$, choose the order-three character χ_2 with $\chi_2(3) = e^{2\pi i/3}$. Since $\chi_2^3 = 1$, uniform averaging gives $\langle \delta\Phi_{\chi_2}^3 \rangle_c = A_{\chi_2}^3$. Define the internal three-point amplitude ratio by

$$\begin{aligned} \mathcal{R}_{3,\chi}(\beta) &:= \frac{|\langle \delta\Phi_{\chi_2}^3 \rangle_c|}{P_{\chi_2}(\beta)^2} \\ &= \frac{1}{|A_{\chi_2}(\beta)|} = \frac{6\zeta(\beta)}{\sqrt{7}\ |L(\beta, \chi_2)|}, \\ \mathcal{R}_{3,\chi}(\infty) &= \frac{6}{\sqrt{7}} = 2.267786838\ldots. \end{aligned}$$

(5.12)

This value belongs to the defined internal complex-phase correlation. Its manifestation in spatial density fluctuations and in three-point statistics of actual radiation records must be obtained through matter formation, propagation, and reception. Section 5.4 gives the relation for calculating this transfer along the same dynamics.

The physical content of the Big Bang is thus developed through a common process: the critical background supplies distinctive energy participation and phase structure, actual exchange determines how it changes, and matter and records determine which changes persist. Beyond the critical region, formation and survival progressively establish the matter composition of the later universe.

5.2 Cooling and Matter Composition

Cooling changes the energy available from the background and the opportunities for matter formation, conversion, and exit. We use the matter species and their phase masses established in the unification theory; cosmology then asks how much of this matter forms during evolution and how much survives. [2]

Comparing thermal and matter energy scales first locates the energy ranges of different processes. For a specified energy E, define $\beta_E^{\mathrm{ref}} = E_*/E$. With the calibration above, the electroweak scale $v_{\mathrm{EW}} = 246.1913739\ldots\,\mathrm{GeV}$ and the nucleon scale $M_B = 0.9366693126\ldots\,\mathrm{GeV}$ of the unification theory under its specified collective preparation give, respectively, [2]

$$\begin{aligned}\beta_{\mathrm{EW}}^{\mathrm{ref}} &\simeq 4.96137\times 10^{16},\\ \beta_{B}^{\mathrm{ref}} &\simeq 1.30403\times 10^{19}.\end{aligned}$$

(5.13)

These values mark where the thermal energy scale $k_{\mathrm{B}}T$ is comparable to the specified matter energy scale E. Formation numbers also depend on channels, exchange strengths, and durations; they can be counted by tracking the formation and survival of successive matter cohorts.

Survival after formation determines whether matter accumulates. Using the formation and first-exit records of Ref. [1], first count a collection with no net transport across its boundary. Let $B_{f,r}$ be the expected number of newly formed members of family f at the end of cycle r, $S_f(R \mid r)$ the probability that this cohort has not exited by the start of cycle R, and $N_{f,0}$ the initial number. Summing cohorts gives

$$\boxed{\begin{aligned}N_{f,R} = N_{f,0}S_f^{(0)}(R)\\ +\textstyle\sum_{r=0}^{R-1} B_{f,r}S_f(R \mid r).\end{aligned}}$$

(5.14)

Survival probabilities are obtained by composing the actual conditional return operations, weighted by the number of carriers in the cohort. Newly formed matter begins to undergo exit from the next cycle onward. Each present carrier is counted once according to its own formation record. The same present total can arise from old matter that persists or from different carriers continually forming and disappearing; these histories have different subsequent responses.

In a realization with scalar survival on a protected carrier in each cycle, let $\epsilon_{f,j}$ be the exit probability and $\Delta\tau_j$ the complete-cycle duration. The probability of persistence from cycle R to cycle $R^{'}$ satisfies

$$\begin{aligned} \Pi_f(R^{'},R) &= \prod_{j=R}^{R^{'}-1}(1-\epsilon_{f,j}) \\ &= e^{-\sum_{j=R}^{R^{'}-1}\Gamma_{f,j}\Delta\tau_j}, \\ \Gamma_{f,j} &= -\frac{\log(1-\epsilon_{f,j})}{\Delta\tau_j}, \\ 1-\Pi_f(R^{'},R) &\leq \sum_{j=R}^{R^{'}-1}\epsilon_{f,j}. \end{aligned}$$

(5.15)

When subsequent cumulative exit, new formation, conversion, and net transport are negligible, the existing composition is approximately preserved and reactions freeze out. Forward and reverse port populations together with full durations determine cumulative change, while the joint recurrence of Section 3.4 supplies each cycle's input.

The net matter–antimatter remnant must be tracked separately. Let Q_B be the net baryon-number operator of the region considered and $\mathcal{B}_r = \mathrm{Tr}(\Omega_r Q_B)$. One complete operation $\mathbb{U}_r$ gives

$$\Delta\mathcal{B}_r = \mathrm{Tr}\left[\Omega_r(\mathbb{U}_r^{\dagger}Q_B\mathbb{U}_r - Q_B)\right].$$

(5.16)

Pair formation can increase the total amount of matter without producing a net difference. If the input is invariant under net-charge reversal and each complete cycle respects this reversal, the final state retains the symmetry and $\mathcal{B}_r = -\mathcal{B}_r = 0$. A nonzero remnant requires a net source generated by the actual preparation or operation and preserved by subsequent exchange. Net change within a region also includes transport and cannot directly be interpreted as failure of global conservation.

The final ratio also involves radiation. For the same comoving cell, take $\eta = \mathcal{B}/N_\gamma$, where N_γ is photon number. Differentiating with respect to common matter time, and writing the net rates as $\mathcal{S}_B = \mathcal{B}^{'}$ and $\mathcal{S}_\gamma = N_\gamma^{'}$, gives

$$\eta^{'} = \frac{\mathcal{S}_B}{N_\gamma} - \eta\frac{\mathcal{S}_\gamma}{N_\gamma}.$$

(5.17)

Even after net baryon number is preserved, later photon production can change this ratio. Formation, survival, net-charge preservation, and radiation jointly determine matter composition.

The magnitude of the baryon ratio can be evaluated using the source-transfer calculation in the unification theory. [24] Under the critical preparation $\beta = 1$, the specified five-layer common transfer has weight $(7n)^{-5\beta}|_{\beta=1} = 7^{-5}n^{-5}$, whose 7-adic part gives $\Gamma_1 = 7^{-5}$. Four of the six characters are complex directions, giving participation factor $\Gamma_2 = 4/6$. With the character convention of Section 5.1, the interference phase comes from

$$\begin{aligned} L(5,\chi_1) &= 7^{-5}\sum_{r=1}^{6}\chi_1(r)\zeta(5,r/7) \\ &= 0.9859999421\ldots \\ &\quad + 0.0295195336\ldots\, i, \\ \Gamma_3 &= \frac{|\mathrm{Im}L(5,\chi_1)|}{|L(5,\chi_1)|} = 0.0299252685\ldots. \end{aligned}$$

(5.18)

The argument 5 of the L-function comes from the five-layer transfer evaluated at criticality, and Γ_3 gives the magnitude of the net asymmetry. Once a source orientation yielding positive net baryon number is chosen, transfer and interference together suppress the net source; equilibrium exchange then determines its baryonic share.

We use the descent-chain balance of Section 16.5 in the sixth version of the unification theory. One descent counts three color addresses and one species address, so $3\mu_q + \mu_\ell = 0$. With common chemical potentials across three generations, mass-vertex equilibrium, hypercharge neutrality, and a common linear thermal-response normalization, the equilibrium relations are [24]

$$\begin{aligned} &\mu_\ell = -3\mu_q, \qquad \mu_u = \mu_q + \mu_H, \\ &\mu_d = \mu_q - \mu_H, \qquad \mu_e = \mu_\ell - \mu_H, \\ &3(\mu_q + 2\mu_u - \mu_d) - 3(\mu_\ell + \mu_e) + 2\mu_H = 0. \end{aligned}$$

(5.19)

Hence $\mu_H = -12\mu_q/7$. Baryon number is $B = 3(2\mu_q + \mu_u + \mu_d) = 12\mu_q$, and lepton number is $L = 3(2\mu_\ell + \mu_e) = -153\mu_q/7$. The fraction of an existing net $B - L$ assigned to baryons is therefore

$$\kappa_B := \frac{B}{B-L} = \frac{28}{79} = 0.35443038\ldots.$$

(5.20)

This allocation coefficient also appears in standard electroweak thermal-equilibrium calculations with three matter generations and one Higgs doublet. [25] We use the balance supplied by the descent chain above, accounting separately for equilibrium allocation and net-source production. Retaining the photon normalization $D_\gamma = N_{\mathrm{vac}}^{\mathrm{eff}}/N_c$ from the same calculation, with $N_{\mathrm{vac}}^{\mathrm{eff}} = 2039.75$ and $N_c = 3$, the factors combine to give

$$
\begin{aligned}
\eta_\star &= \frac{\Gamma_1\Gamma_2\Gamma_3\kappa_B}{D_\gamma},\\
D_\gamma &= \frac{2039.75}{3},\\
\eta_\star &= 6.1877370\ldots\times10^{-10}\\
&\simeq 6.19\times10^{-10}.
\end{aligned}
\tag{5.21}
$$

This corresponds to about 0.619 net baryons per billion photons. Source transfer, interference associated with the complex-character directions, and equilibrium allocation jointly determine the net remnant; D_γ converts it to a photon-number ratio according to the stated counting correspondence.

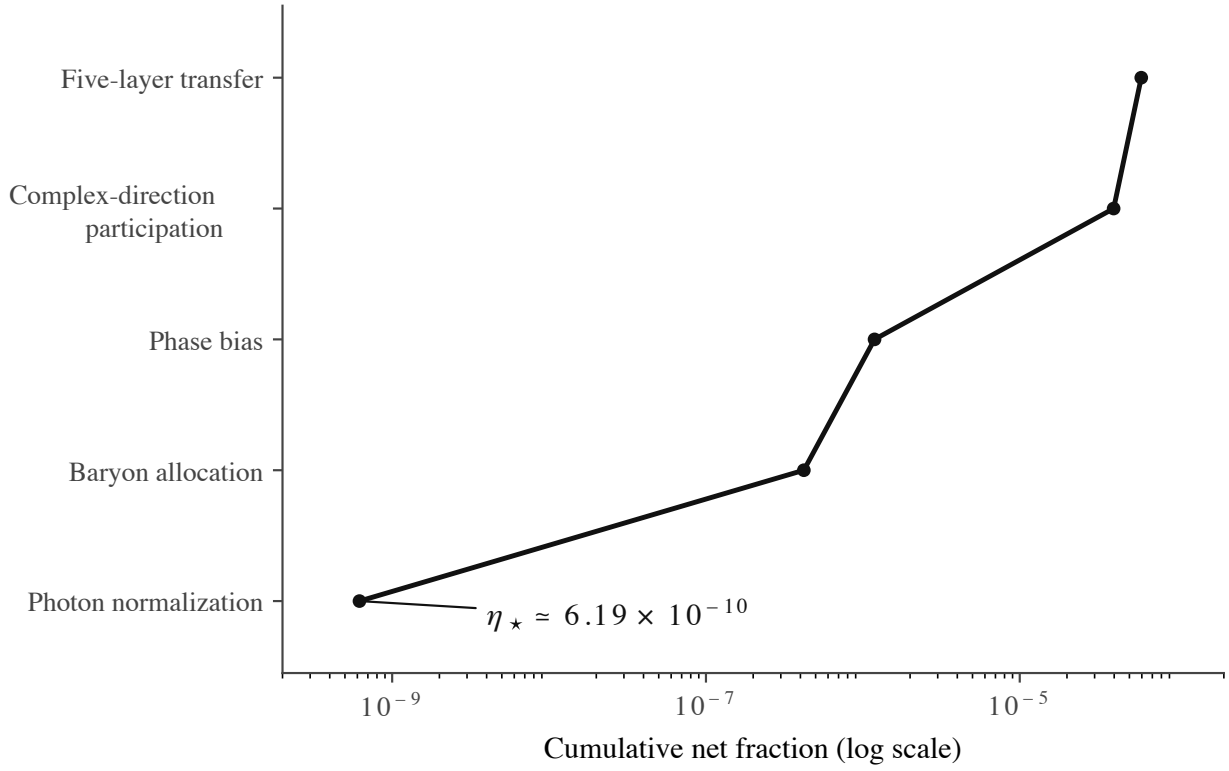


Figure 5-2. Evaluation of the net baryon ratio. Successive points incorporate the specified transfer, participation, phase, allocation, and photon-normalization factors, showing how the net fraction reaches the 10^{-10} scale. The horizontal axis gives the cumulative fraction and the vertical sequence lists evaluation steps; these factors are not treated as independent cosmic epochs.

The resulting source must survive subsequent evolution to become a later abundance. In a window with small net asymmetry where linear washout applies, let s_r be net baryon number newly produced in cycle r and w_r the fraction of an existing net difference erased in that cycle. Then

$$\begin{aligned}\mathcal{B}_{r+1} &= (1-w_r)\mathcal{B}_r + s_r,\\ \mathcal{B}_R &= \mathcal{B}_0 \prod_{j=0}^{R-1}(1-w_j)\\ &\quad + \sum_{r=0}^{R-1} s_r \prod_{j=r+1}^{R-1}(1-w_j).\end{aligned}$$

(5.22)

Each cycle's net source is counted with its subsequent retained fraction. Once source and washout realize the normalization above, conservation of net baryon number and comoving photon number carries $\eta_\star$ into the late era. Continuing washout, photon production, and entropy transfer update the ratio through number balances. Comparison with primordial deuterium and microwave-background observations additionally requires calculating how nuclear reactions and radiation propagation transmit these quantities. [21,26]

Surviving matter continues to contact and alter the common background. Changes in composition therefore enter subsequent maintenance and propagation as actual participants in cosmological scale evolution.

5.3 Cosmic Expansion and Redshift

Background change can affect both matter phase and light propagation. Comparing them under the common calibration of Chapter 4 gives the scale $a = (F/G)/(F_0/G_0)$. Let a prime denote differentiation with respect to matter time τ, and a dot differentiation with respect to the complete process parameter t. Then

$$\boxed{\begin{aligned}H_{\mathrm{geom}} &:= \frac{a'}{a},\\ H_{\mathrm{geom}} &= \frac{1}{F}\left(\frac{\dot F}{F} - \frac{\dot G}{G}\right).\end{aligned}}$$

(5.23)

Sustained expansion requires this difference to remain positive over the readable interval. When common activity time gives both rates the same multiplier, the scale stays fixed; growth requires the established physical operations to affect them differently.

Resource restoration provides an explicit realization. The examples in this section use energy unit E_* and time unit $\hbar/E_*$. Gaps, couplings, and durations without explicit units denote their numerical values in these units; factors of E_* or $\hbar$ restore physical quantities. A matter source first writes a bias into the actual maintenance resources, giving initial availability $h_0 = 1/2 - b$, with $b = \sin(2\theta\lambda_s)/2$ and $0 < 2\theta\lambda_s < \pi/2$. Each resource C_j then undergoes the same exchange with a prepared half-occupied auxiliary A_j. The stage begins after writing and ends before the next source contact; preparation time

and its records remain included in the whole process. There are M pairs of actual participants, with input $\rho_{CA}^{\otimes M}$, where $\rho_{CA} = \mathrm{diag}(h_0, 1-h_0) \otimes I/2$. The operation containing the two selected matter clocks and resource restoration is

$$
\begin{aligned}
H &= S \otimes \overline{h} + \omega_R \textstyle\sum_{j=1}^{M} X_j + K_\gamma, \\
\overline{h} &= \tfrac{1}{M} \textstyle\sum_j (I - N_{C_j}), \\
X_j &= |01\rangle\langle 10| + |10\rangle\langle 01|.
\end{aligned}
$$

(5.24)

Here S is the actual maintenance generator of the selected matter, and K_γ is the established photon propagation operator. The matter input has readable coherence and is initially uncorrelated with the resource pairs; old outputs and preparation records from writing are retained. Resources gate only the maintenance term, so photon free flight in this stage remains generated by K_γ, giving $G = 1$ within the original long-wavelength calibration range. This is an output of the operation; matter back-action on resources remains included in $S\overline{h}$ throughout.

For two distinct eigenvalues λ_a, λ_b of S, let $h = I - N_C$. Conditional propagation of a single pair is $U_a = e^{-it(\omega_R X + \lambda_a h/M)}$. Define $R_{ab} = U_a \rho_{CA} U_b^\dagger$, $\gamma_{ab} = \mathrm{Tr} R_{ab}$, and $n_{ab} = \mathrm{Tr}(h R_{ab})$. Initial independence gives the matter coherence multiplier $\Gamma_{ab} = \gamma_{ab}^M$. Substitution into the cross-block clock rate of Section 4.4, normalized by its input frequency, gives on a nonzero-coherence window

$$
F_{ab}(t) = -\frac{\partial_t \arg \Gamma_{ab}}{(\lambda_a - \lambda_b) h_0} = \frac{\mathrm{Re}(n_{ab}/\gamma_{ab})}{h_0}.
$$

(5.25)

The clock rate is therefore obtained from actual coherence. A comparison trajectory for the same resource exchange clarifies the restoration effect: without the matter load, availability is $h_{\mathrm{ref}}(t) = 1/2 - b\cos^2(\omega_R t)$. Over finite elapsed time and within the stated large-capacity range, exact conditional propagation gives a controlled approximation to the common response:

$$
\begin{aligned}
F_{ab}(t) &\simeq \frac{h_{\mathrm{ref}}(t)}{h_0}, \qquad G(t) = 1, \\
a_{\mathrm{ref}}(t) &= \frac{1/2 - b\cos^2(\omega_R t)}{h_0}, \\
\dot{h}_{\mathrm{ref}}(t) &= b\omega_R \sin(2\omega_R t).
\end{aligned}
$$

(5.26)

During the rising part of resource restoration, the matter phase accelerates relative to light propagation and the common comparison scale grows. We now test this interval directly using derivatives of the exact coherence.

For example, take $\theta = 0.4$, $\lambda_s = 0.7$, $\omega_R = 3$, and the joint maintenance generator $S = 0.84B_1 + 1.08B_2$ for the two materials, where each B_i has eigenvalues ± 1 and acts on its respective material. Set $M = 256$. Over $\pi/(8\omega_R) \le t \le 3\pi/(8\omega_R)$, the clock-rate derivatives for all six cross coherences satisfy $\dot{F}_{ab} > 2.1784$, and all coherence magnitudes $|\Gamma_{ab}|$ exceed 0.9911. Each clock rate differs from the common reference by less than 0.051 throughout the interval, while the comparison scale grows from 1.1659 to 1.9671.

Endpoint readings of the six coherences under the same exact propagation can also be compared directly with the common reference. At the two displayed growth endpoints, every actual clock rate differs from the common curve by less than 1.8×10^{-6}, with high visibility retained. These endpoint checks and the analytic derivative bounds over the full interval jointly support finite growth.

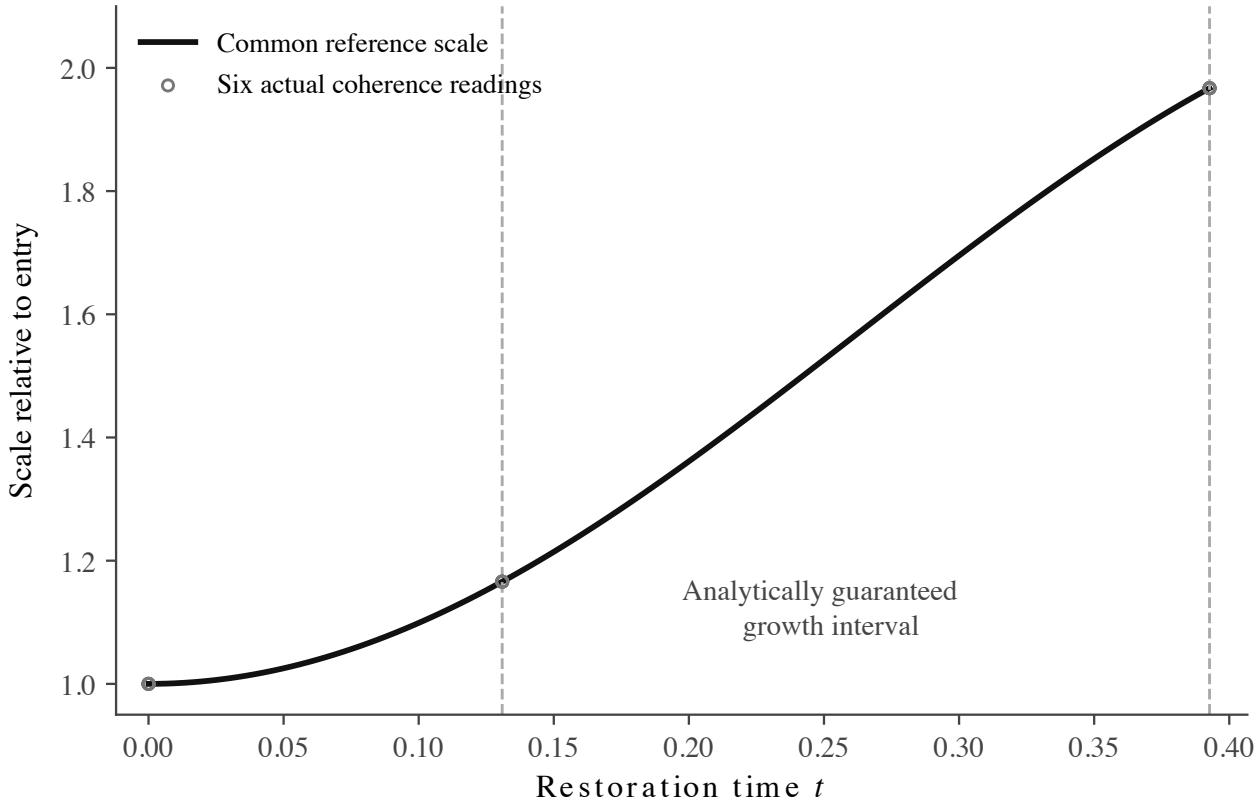


Figure 5-3. Scale growth during restoration. The continuous curve is the common reference a_{ref}; circles are six coherence readings at the endpoints under the same propagation, nearly coincident at the displayed resolution. Dashed lines mark the interval of analytically guaranteed growth. Elapsed time on the horizontal axis is measured in $\hbar/E_*$.

Where the common matter calibration of Chapter 4 and uniform, nearly nondispersive propagation apply, the optical manifestation of scale change follows from emission and reception of the same record. Neighboring emission events at fixed reference distance satisfy $d\tau_r/d\tau_e = a_r/a_e$. If the outgoing phase is conveyed by the light signal, the chain rule gives

$$\omega_r = \omega_e \frac{d\tau_e}{d\tau_r},$$
$$\boxed{1 + z = \frac{\omega_e}{\omega_r} = \frac{a_r}{a_e}.}$$

(5.27)

Emission and reception frequencies use the same matter calibration at both ends. The arrival relation already includes the actual propagation delay. When neighboring wave packets and the instrument resolve this phase, a growing scale appears as a longer reception interval and a lower received frequency.

A material spectral line additionally requires emission to match a transition. For the established matter–photon exchange under a controlled scalar comparison propagator, let $E(t)$ be the transition gap of the emitter, ω_k the photon mode frequency, and κ_k a fixed coupling weight. In an emission window centered at t_e with length L_e, the first-order outgoing amplitude is

$$\begin{aligned} \phi_e(s) &= \int_0^s E(t_e - L_e/2 + u)\, du, \\ A_k^{(1)} &= -ig_{\mathrm{em}} \kappa_k e^{-i\omega_k L_e} \\ &\quad \times \int_0^{L_e} e^{i[\omega_k s - \phi_e(s)]}\, ds. \end{aligned}$$

(5.28)

In a slowly varying window, E can be approximated by its central value, and the integral gives a finite-width spectrum centered at $\omega_k = E_e$. Later matter of the same kind supplies a local gap E_r, and the receiver resolves the frequencies of arriving modes. Choose a separate set of restoration parameters suitable for a slowly varying spectral line. Free flight preserves mode frequencies, while both endpoint gaps follow h_{ref}, giving $1 + z \simeq E_r/E_e = a_r/a_e$. The actual loaded source retains the environmental two-time kernel of Section 4.5. Its difference from the scalar comparison, together with finite-window, resource, and reception errors, controls line resolution; switching work is counted as in Section 3.5.

This slowly varying spectral-line process also has explicit values. Keep the same bias $b = 0.265593099\ldots$, and take restoration frequency $\omega_R = 10^{-5}$ and participation capacity $M = 10^{20}$. Use equal-charge logical levels $\lambda_e = 1.92$, $\lambda_g = 0.24$ of the specified photon coupling, with ungated gap 1.68. Set emission coupling $g_{\mathrm{em}} = 0.005$, reception coupling $g_D = 0.02$, and both window lengths to 60. This is the stated material-transition realization; no specific atomic spectral line has yet been assigned to it. The window centers are $t_e = \pi/(8\omega_R)$ and $t_r = 3\pi/(8\omega_R)$, giving effective material gaps

$$E_e = 0.459147545\ldots, \qquad E_r = 0.774656049\ldots.$$

(5.29)

Energy and time use the common units of this complete process. The photon sector uses 401 modes on an 8191-site position ring, with $k_n = 2\pi n/8191$, $n = 50, \ldots, 450$, and frequencies $\omega_n = 3\sin k_n$. All modes couple with equal amplitude, and the connections remain fixed. The effective optical calculation

retaining the slow variation of $h_{\mathrm{ref}}(t)$ gives the discrete peak of the earlier outgoing spectrum at $\omega_{\mathrm{peak}} = 0.458447079\ldots$, close to the contemporaneous material transition. Free flight preserves mode frequencies, whereas the later local material supplies reference E_r. The spectral-center comparison and the finite-band peak of the effective spectrum therefore give, respectively,

$$\begin{aligned} \frac{E_e}{E_r} &= 0.59271149\ldots, \\ z_{\mathrm{center}} &:= \frac{E_r}{E_e} - 1 = 0.68716148\ldots, \\ \frac{\omega_{\mathrm{peak}}}{E_r} &= 0.59180727\ldots. \end{aligned} \tag{5.30}$$

The small difference between center and peak comes from the finite emission window, discrete band, and slow variation within the window. Bounds on finite-resource error give the conclusion for the actual joint output: the emission probability is at least 0.00914, and, conditional on emission, at least approximately 87.67% of the photon probability lies in

$$0.47653 \le r := \frac{\omega}{E_r} \le 0.70889. \tag{5.31}$$

The entire band lies below the local reference $r = 1$, retaining redshift at finite resolution. Reception uses a mode-resolving terminal instrument, whose tuning and readout errors are included in the complete process. The discrete peak remains a result of the effective calculation. The figure uses a first-order frozen-window reference spectrum to display the shift, with the difference from the actual output controlled by the bounds above. This slowly varying configuration and the earlier $M = 256$ growth example are separately prepared, illustrating the same restoration mechanism in different readable windows.

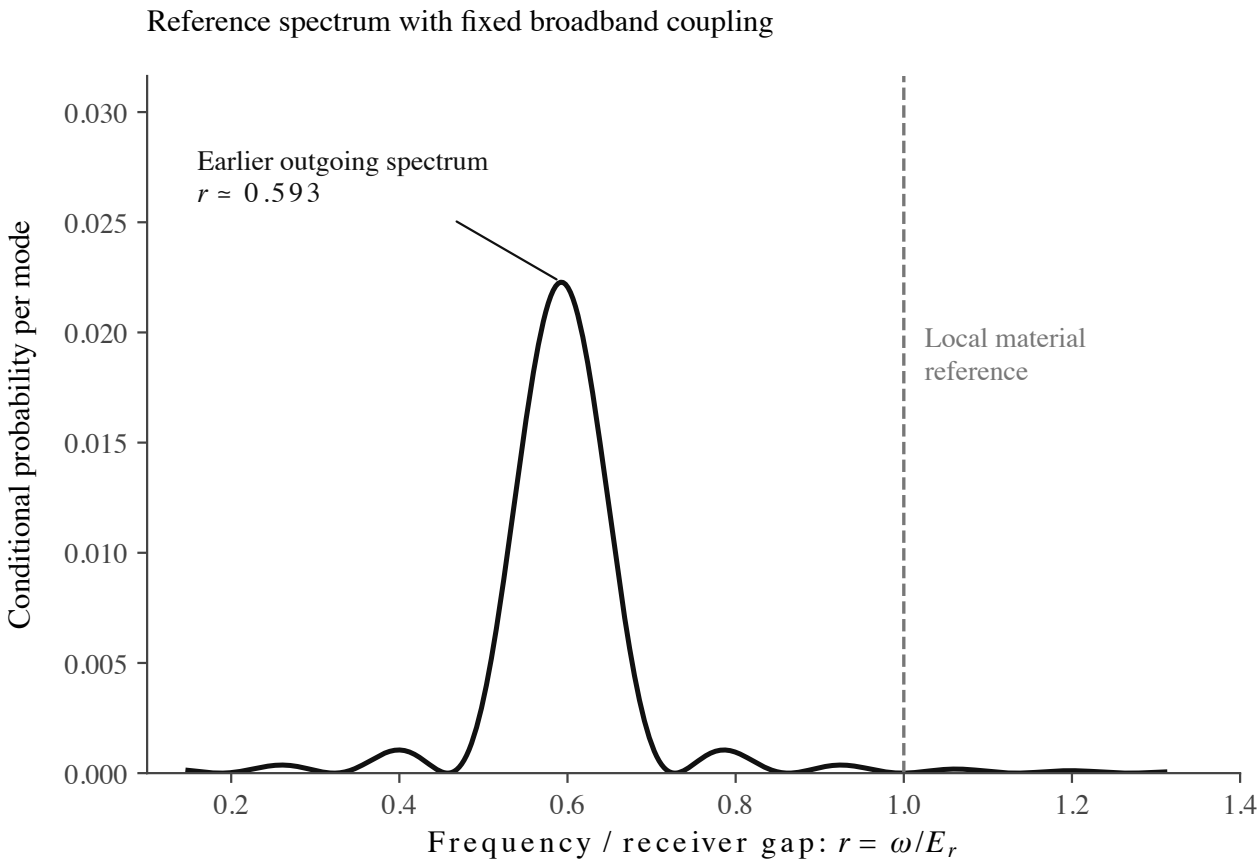


Figure 5-4. Comparison of an earlier outgoing spectrum with a later matter reference. The horizontal axis is photon frequency divided by the gap of the same kind of matter at reception. The dashed line $r = 1$ marks the local reference; the earlier spectrum is concentrated near $r \simeq 0.593$. The curve is the stated finite-window reference spectrum, and its vertical normalization is conditional on an emitted photon; it does not imply unit emission probability.

The preceding growth and narrow-line redshift occur during restoration after source writing. Extending the process makes writing and restoration jointly determine the subsequent bias. Even before the source is reconnected, coherent return in restoration itself can reverse growth. Take symmetric matter branches $\pm\lambda$, and let $q = \lambda/(2M)$, $\kappa = (\omega_R^2 + q^2)^{1/2}$. At return times $t_m = m\pi/\kappa$ on the original optical branch, set $\alpha = qt_m$. Where coherence is nonzero, the phase rate satisfies

$$\frac{F}{G} - 1 = -\frac{2b(1+2b)\sin^2\alpha}{1-(1-4b^2)\sin^2\alpha}.$$

(5.32)

Resource occupation then returns to its input value, yet the matter phase rate can fall below its input value. The history stored in coherence makes occupation and clock rate return differently, motivating sustained service.

A sufficient condition for preserving the bias follows directly from the joint generator. Suppose that after complete service the state is supported on an invariant subspace P, with $\bar{h}P = h_* P$ and $HP = (E_* h_* S + H_{\mathrm{rest}} + K_\gamma)P$, where the latter two terms act only on participants other than the compared matter spectral labels. Then

$$Z_{ab}(t) = Z_{ab}(t_1) e^{-iE_*(\lambda_a - \lambda_b)h_*(t-t_1)/\hbar}, \qquad \frac{F_{ab}}{G} = \frac{h_*}{h_0}.$$

(5.33)

This inherits $G = 1$ on the original invariant optical branch. The conditions $h_* > h_0$ and nonzero initial coherence give a persistent multiplier above its input value. Old stores may remain correlated with matter, since the local propagator above acts directly on the complete successor. Finite coherent-store compensation, transfer of the old resource state into a blank archive, and protective connections after service can realize this condition. A constant multiplier has zero growth rate: maintaining a larger scale and continuing to grow therefore have different dynamical requirements.

An internal program can execute these energy-conserving services in a separately specified preparation. Let Q_i be the complete unitary service on resource i. The first M edges perform archiving and signed store compensation, and $Q_i = I$ thereafter. Here $H_0 = E_* S\bar{h} + H_W + K_\gamma$ contains matter maintenance, finite stores, and the original light propagation, with additional nonparticipating spectator terms. The services satisfy $[Q_i, H_0] = 0$, and the autonomous generator is

$$H_{\mathrm{aut}} = H_0 + \hbar\gamma \sum_{n \geq 0} \left(|n+1\rangle\langle n| \otimes Q_{n+1} + \mathrm{h.c.} \right).$$

(5.34)

The program retains every reverse edge, so returns to the old region after service enter the actual phase. This realization prepares finite archives, records, and coherent stores, and additionally requires an unbounded accessible set of program positions. It is prepared separately from the preceding continuous coherent exchange. Transforming the program to half-line propagation with the history unitary $W_n = Q_n \cdots Q_1$, and retaining overlaps of conditional store translations, gives matter coherence of the form

$$\Gamma_{ab}(t) = e^{-i\nu_{ab}t}[q_{ab} + e_{ab}(t)], \qquad \nu_{ab} = \frac{E_*(\lambda_a - \lambda_b)}{\hbar}.$$

(5.35)

Here $q_{ab} > 0$ is the store overlap after service, and e_{ab} retains all corrections from returns to the old region. The half-line propagation kernel and its actual derivative separately bound the late-time decay of $e_{ab}, \dot{e}_{ab}$. Direct differentiation of the phase above gives

$$\left|F_{ab}-\frac{1}{h_0}\right|\le\frac{|e_{ab}|}{|\nu_{ab}|h_0(q_{ab}-|e_{ab}|)},\qquad |e_{ab}|<q_{ab}.$$

(5.36)

This bound controls both readability and phase rate. Use h_0, $M=256$, and the two materials of the earlier growth example. Set $\gamma=E_*/\hbar$, take store spacing $0.12E_*/M$, and prepare equal amplitudes on $64M$ levels with safety margins of $16M$ levels on each side. All required translations stay within the finite store. Measuring time from completion of preparation, the clocks obey $F_1/G\ge 3.6035$ and $F_2/G\ge 3.6983$ for all $t\ge 1024\hbar/E_*$, with nonzero coherence throughout; both tend to $h_0^{-1}=4.266086\ldots$. A small mismatch remains between the clocks, and each growth rate continues to change sign at arbitrarily late times. This service preserves multipliers above entry, producing clock-rate multipliers that approach a common plateau.

The autonomous service retains conditional correlations among stores, resources, and program, which also enter late reception of early emitted light. Denote the two original loaded materials by A,D, and use the same initial matrix in their respective $(+,-)$ bases:

$$\rho_A=\rho_D=\begin{pmatrix}0.65 & 0.27+0.08i\\ 0.27-0.08i & 0.35\end{pmatrix},\qquad \rho_{AD}=\rho_A\otimes\rho_D.$$

(5.37)

The emission port is $L_A=\sigma_A^-\otimes I_D$, including both equal-charge transitions $++\to -+$ and $+-\to --$. The reception port maps D^- together with one photon to D^+ and a click record. Both ends use all 4095 positive-frequency modes of the 8191-site position ring, with $k=2\pi n/8191$, $1\le n\le 4095$, $\omega_k=3\sin k$, and coupling weights $\kappa_k=1/\sqrt{4095}$ and $\kappa_{\chi,k}=e^{-ik\chi}/\sqrt{4095}$, respectively. A ready click record and blank readout bit are included in the initial preparation. The participating reception weight in the original input is $p_{\mathrm{acc}}=0.65\times 0.35$; the calculation retains coherence of the complete initial state and its other branches.

Choose early emission over $0\le t\le 60$ with $g_S=0.005$. Retain the emitter's joint correlations with stores, resources, and program, propagate freely to $t_r=10^8$, then let material D receive at $\chi=256$ for duration 60 with $g_D=0.02$, followed by readout lasting $\pi/6$. These durations are in $\hbar/E_*$; the complete duration also includes initial preparation, and the background at arrival is not reprepared.

Using the reception gap E_D as the instrument setting, the combined error of one emission insertion and the complete reception comparison is bounded by $\varepsilon\le 1.026\times 10^{-4}$. The probability envelope in Chapter 4 gives approximately the following actual outputs for two settings:

$$\begin{aligned}E_D=1.05E_*:&\quad 1.68\times 10^{-8}<q<1.13\times 10^{-7},\\ E_D=1.68E_*:&\quad 0\le q<1.52\times 10^{-8}.\end{aligned}$$

(5.38)

The same early correlated light thus produces nonzero late reception records and distinguishes these settings. If the reception gap is adjusted by δE_*, conservation of $N_D - P_{\text{click}}$ gives net tuning work $-\delta q E_*$. This terminal response retains intermode phases, program returns, and the finite frequency band. Spectral-peak inversion and first-arrival resolution are further, distinct readings.

These calculations realize matter phases, narrow-line frequency shifts, and correlated light reception as concrete events. Finite restoration and autonomous service yield results under different preparations. A cosmological distance–redshift relation further requires connecting matter, background supply, and full time accounting across epochs. This relation is also what supernova and baryon acoustic oscillation observations test. [27,28]

5.4 Matter Distribution and Gravitational Structure

Matter distribution first concerns the components retained in the common background and their amounts. Net baryons are counted through the formation and retention processes of Section 5.2. Persistent matter lacking effective exchange with visible radiation but possessing a net gravitational response can form a dark component in the corresponding observations. The unification theory gives component values for $p = 7$, [24] which can be connected to formation–exit processes in finite reductions. The dark component must also be tested through its spatial gravitational response; separation of lensing mass from visible hot gas in galaxy clusters supplies such an observational constraint. [29]

Take the five nontrivial directions modulo 7 as a closed participating sector. Let return preserve directions, let restoration use Haar shell fractions $g_j = |S_j|/7^K$, and set the local thermal parameter to $\beta_{\text{loc}} = 1$. This parameter describes the finite exchange environment. Write f_ν for the per-cycle formation coefficient of direction ν, $\epsilon_\nu > 0$ for its exit probability, and y for the background population. With fixed preparation parameters and formation–exit balance established,

$$\epsilon_\nu x_{\nu,*} = y_* f_\nu, \qquad \frac{x_{\nu,*}}{y_*} = \frac{f_\nu}{\epsilon_\nu}.$$

(5.39)

Of the five directions, χ_1, χ_5 have full order. Couple these to the terminal zero shell, and the other three to the ordinary zero-energy shell-average state; each block uses the conserving thermal exchange established in state-space dynamics. [1] The ordinary zero-energy shell has Haar fraction $u = 6/7$, and equal-energy endpoints give $f_\nu/\epsilon_\nu = u$. The terminal fraction is 7^{-K}, while the detailed-balance factor for gap $K\log 7$ is 7^K; their product gives $f_\nu/\epsilon_\nu = 1$. Hence

$$\frac{x_{\nu,*}}{y_* u} = \begin{cases} 1, & \nu = 2,3,4, \\ 7/6, & \nu = 1,5, \end{cases}$$

(5.40)

The ratio of total directional occupation to the leading unit $y_* u$ is

$$\boxed{R_{\mathrm{occ}} = 3 + 2\frac{7}{6} = \frac{16}{3}.}$$

(5.41)

The enhancement arises from actual stationary occupation. The common strength of formation and exit cancels from this ratio, although the time required to reach stationarity still depends on those strengths and on depth. When each direction carries the same physical energy per unit occupation, let ρ_{unit} be the physical density of one leading occupation unit in the common volume. Then

$$\frac{\bar{\rho}_d}{\bar{\rho}_b} = \Lambda_{\mathrm{comp}} R_{\mathrm{occ}}, \qquad \Lambda_{\mathrm{comp}} := \frac{\rho_{\mathrm{unit}}}{\bar{\rho}_b}.$$

(5.42)

With the common normalization $\Lambda_{\mathrm{comp}} = 1$ used in this component evaluation, the physical dark-component-to-baryon density ratio is $\bar{\rho}_d/\bar{\rho}_b = 16/3$. The baryon reference is supplied by actual baryon formation, while the shell-average background participates only in formation and exit. The correction for the full-order directions is already included in occupation and is not applied again to physical energy. During a stage containing only these two kinds of long-lived nonrelativistic matter, their fractions of total matter are

$$\boxed{\begin{aligned} f_b &= \frac{1}{1 + 16/3} = \frac{3}{19} \simeq 15.79\%, \\ f_d &= \frac{16/3}{1 + 16/3} = \frac{16}{19} \simeq 84.21\%. \end{aligned}}$$

(5.43)

Each unit of baryon density thus corresponds to about 5.33 units of dark-component density. Percentages use the total of these two matter components as denominator. When subsequent net formation, conversion, and exit are negligible, the mean ratio remains fixed while local position distributions continue to evolve.

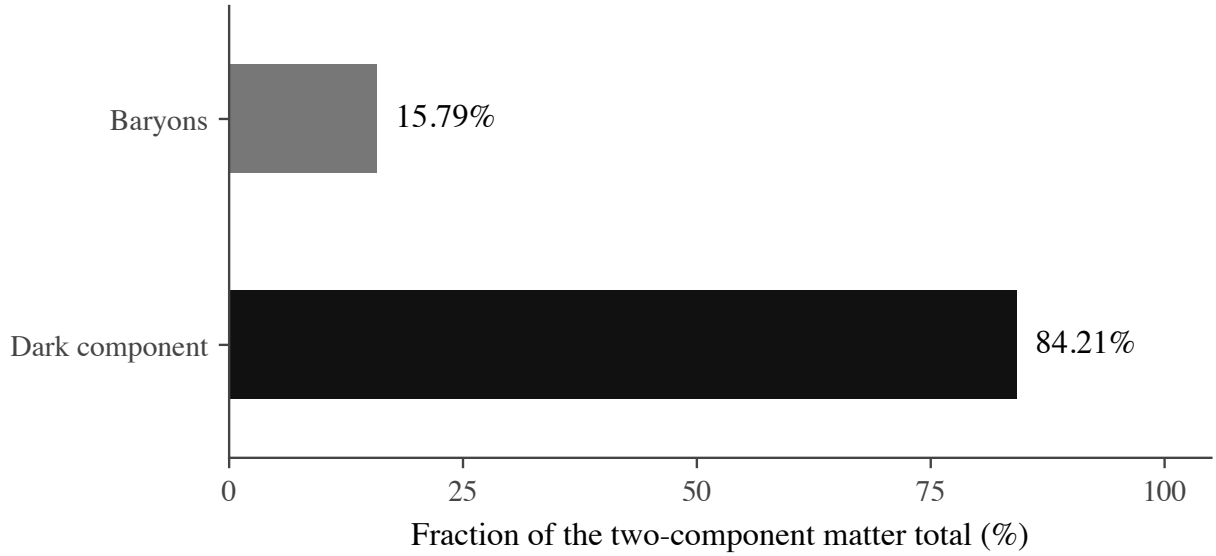


Figure 5-5. Abundances of two matter components. Return through the terminal shell enhances the stationary occupation of the full-order directions, giving $R_{\mathrm{occ}} = 16/3$. Under the common normalization above, baryons and the dark component account for 15.79% and 84.21%, respectively, of their combined total.

Scale change also affects mean matter and radiation distributions. Consider a uniform, isotropic, long-wavelength range with common calibration, and a spatial realization in which the same scale a calibrates physical lengths in all three directions, together with the corresponding volume measure. Using the principal mass–motion operator of the unification theory, let $\boldsymbol{\kappa}$ be a fixed comoving wavevector. The positive-energy spectrum and physical volume of a comoving cell are [2]

$$\epsilon_{f,\boldsymbol{\kappa}} = \sqrt{m_f^2 + \frac{\hbar^2 c_0^2 |\boldsymbol{\kappa}|^2}{a^2}}, \qquad V = a^3 V_\chi.$$

(5.44)

Here $\hbar$ is restored, m_f is the rest-phase spectral value measured in energy units, and $\boldsymbol{\kappa}$ is the comoving wavevector. The quantity m_f is read in common matter time; its change is an actual change in mass ratio relative to the reference matter. If no formation, exit, or exchange occurs during the stage and comoving-mode occupations are preserved, the energy of each massless mode scales as a^{-1}. Dividing by volume gives the radiation response a^{-4}. For a protected positive-mass branch, additionally assume preserved number and relative mass, with kinetic energy much smaller than rest energy. Rest mass then gives the leading energy, so density changes only through volume dilution. Therefore

$$\boxed{e_\gamma a^4 = \text{constant}, \qquad e_{\mathrm{m}} a^3 \simeq \text{constant}.}$$

(5.45)

These dilution relations also apply to nonthermal distributions. Contributions from formation, exchange, and drifting mass ratios enter the energy balance as in Section 3.5, so matter density evolves with both scale and composition.

Beyond mean dilution, actual distributions contain spatial differences and correlations. For a persistent carrier, let P_x be the position projection, $p_x = \mathrm{Tr}(\Omega P_x)$, and H_{mov} its actual motion operator. The joint equation of Chapter 3 gives the position probability current:

$$J_{y\to x} = i\mathrm{Tr}\left[\Omega(P_y H_{\mathrm{mov}} P_x - P_x H_{\mathrm{mov}} P_y)\right],$$
$$\dot{p}_x\big|_{\mathrm{mov}} = \sum_y J_{y\to x}, \qquad J_{y\to x} = -J_{x\to y}.$$

(5.46)

Position coherence determines probability current, while formation and exit change numbers as in Section 5.2. The directional differences of Section 5.1 enter spatial statistics through these operations. Let O_x be a self-adjoint observable at position x, such as matter number or calibrated local density. Include formation, motion, background interactions, and records in the propagator $\mathbb{U}(t, t_0)$, and write

$$O_x(t) = \mathbb{U}(t,t_0)^{\dagger} O_x \mathbb{U}(t,t_0),$$
$$\delta O_x(t) = O_x(t) - \mathrm{Tr}[\Omega(t_0) O_x(t)] I.$$

(5.47)

The symmetrized connected correlation between positions can then be obtained from the same initial state:

$$\boxed{C_O(x,y;t) = \frac{1}{2}\mathrm{Tr}\left[\Omega(t_0)\{\delta O_x(t), \delta O_y(t)\}\right].}$$

(5.48)

This expression retains both input differences and dynamically generated correlations. For mutually commuting matter-number observables, it is $\langle \widehat{n}_x \widehat{n}_y\rangle - \langle \widehat{n}_x\rangle\langle \widehat{n}_y\rangle$. When the mean state and correlation are translation invariant, set $r = x - y$ and take the spatial Fourier transform:

$$P_O(k;t) = \sum_r e^{-ik\cdot r} C_O(r;t).$$

(5.49)

The internal directional intensity P_χ is determined by the character decomposition of a phase observable; the spatial power spectrum P_O is determined by actual position readings. Intervening formation, motion, and reception operations supply the transfer relation between them. Thus 7/36 and the internal three-point amplitude ratio remain quantitative input-correlation results, while the shape of the spatial spectrum and observed three-point statistics must still be obtained through this transfer.

Differences in distribution in turn alter the common background through matter sources. Use the source instrument of Chapter 3 during a protected readout stage that preserves position. Let W_y project onto positions where the current matter contacts background slot y, and set $\xi_y = N_y - I/2$. Then

$$\Delta\langle \xi_y \rangle = \varepsilon \left\langle W_y \left[\frac{1}{2} \sin(2\theta F_f) - \xi_y \right] \right\rangle .$$

(5.50)

Matter position, internal state, and local background jointly determine the magnitude and direction of net writing.

Spatial transport of the background then conveys this change to other positions. In a realization with a fixed eigenstate source, specified background transport, and closed mean evolution, let $\boldsymbol{\xi}$ be the resource bias, D the diagonal matrix of replacement fractions, $\boldsymbol{b}$ the source bias, and R the actual transport matrix. Successive cycles give

$$\boldsymbol{\xi}_{n+1} = R\big[(I - D)\boldsymbol{\xi}_n + D\boldsymbol{b}\big].$$

(5.51)

If $R = I - \eta_R L$, where L is the Laplacian of the specified position graph, denote the refreshed stationary bias by $\boldsymbol{\xi}^+ = (I - D)\boldsymbol{\xi} + D\boldsymbol{b}$. Then $\eta_R L \boldsymbol{\xi}^+ = D(\boldsymbol{b} - \boldsymbol{\xi})$, so actual net writing supplies the field source. For an infinite uniform three-dimensional nearest-neighbor graph, a localized source, and bias tending to zero at infinity, normalization to unit lattice spacing gives $\widehat{L}(\boldsymbol{k}) = |\boldsymbol{k}|^2 + O(|\boldsymbol{k}|^4)$ at small wavevectors. Outside the source, with only conservative transport and at distances large compared with the spacing, the stationary Green kernel has leading term $1/(4\pi r)$. At finite time this form applies only where transport has established a quasistationary response. A finite closed graph must also retain the constant zero mode, with total stationary net writing equal to zero. Actual spacing, contact fractions, and transport determine the physical coefficient; any local refresh or loss remains in the corresponding operation.

Test matter reads this background through its own maintenance. In the established window of controlled averaging, positive energy, low momentum, weak field, and slowly varying background, the change in the local rest generator relative to the half-occupation reference is $\delta h_{\mathrm{rest}} = -2\xi h_{\mathrm{rest},0}$. With the calibration $\hbar = c_0 = 1$ from the same process, we can write

$$\Phi = -2\xi, \qquad V_f = m_f \Phi, \qquad \boldsymbol{f}_f = -m_f \boldsymbol{\nabla}\Phi.$$

(5.52)

This local weak-field reading follows the maintenance and motion mechanism of the unification theory; the common cosmic clock rate is still calculated from actual coherence as in Chapter 4. If the positive bias created by a net source decreases with distance, Φ is lower near the source and test matter experiences an inward action. The resulting position probability current changes the matter

distribution, which then enters the source action again, producing feedback toward gravitational aggregation. Exchange, transport, motion, and actual composition jointly determine its growth rate and spatial extent.

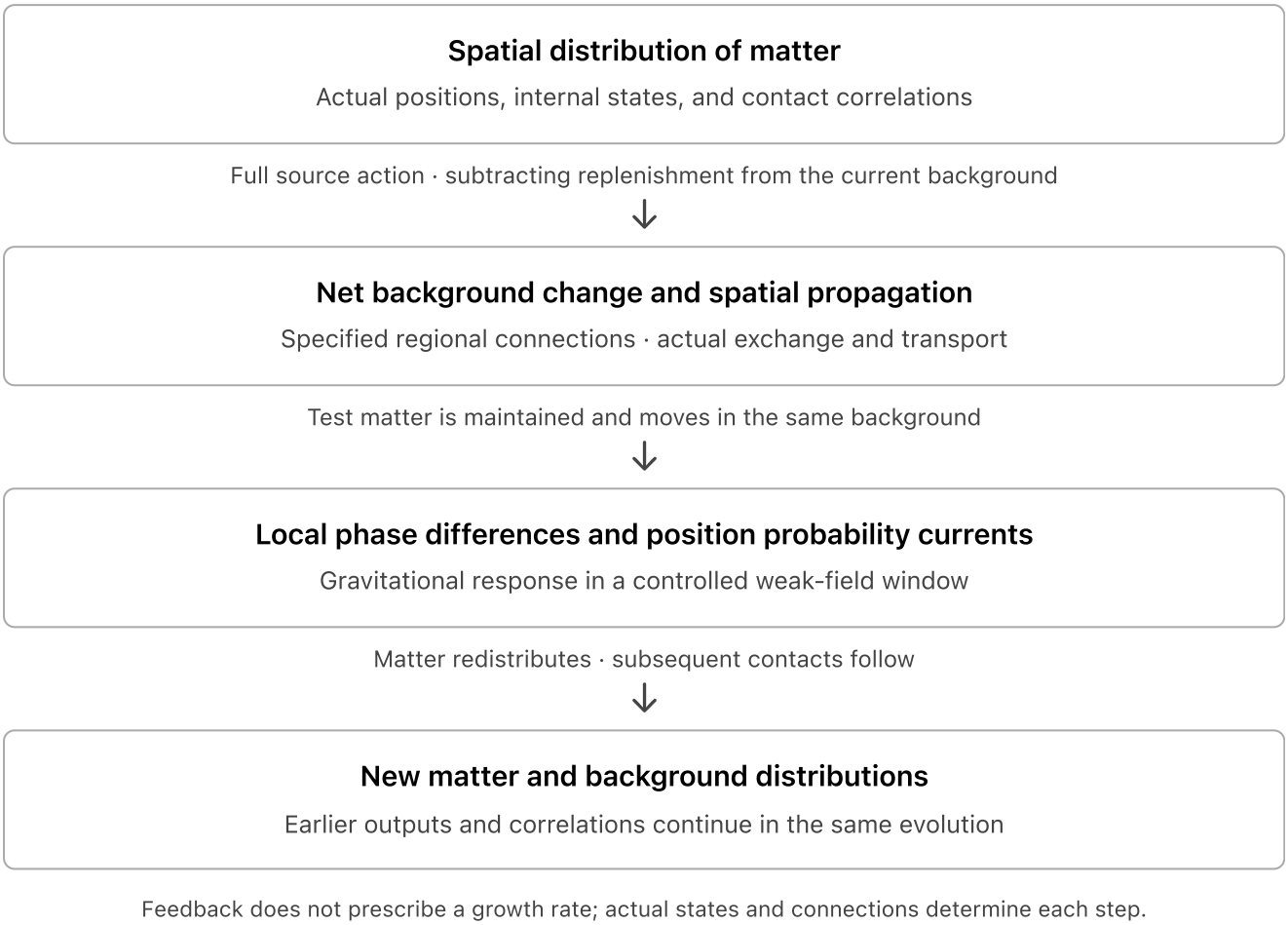


Figure 5-6. Spatial distribution and gravitational feedback. Matter's position distribution changes the net background response. Background propagation affects the maintenance and motion of other matter, and position probability currents alter the distribution again. Existing background and position correlations evolve together with this feedback.

Mean component ratios follow from formation and return, while position, matter state, and background response continue to drive local aggregation. Different prime channels obtain corresponding readings of the same whole, and each matter carrier is counted once in the gravitational source.

5.5 Late Evolution and the Final State of the Universe

The late state of the universe depends on how long these processes can continue. Formation may weaken while existing matter persists. Resource restoration can change current contact conditions, while earlier background changes and records are retained. Cooling, disappearance of matter, and scale growth therefore have their own dynamical conditions.

First consider persistent matter. In the scalar-survival regime of Section 5.2, if $0 \le \epsilon_{f,j} < 1$, the infinite-survival probability from cycle R is

$$\Pi_f(\infty, R) = \exp\left[- \sum_{j=R}^{\infty} \Gamma_{f,j} \Delta\tau_j \right].$$

(5.53)

Along a sequence of operations that can be continued, divergent cumulative exit implies eventual exit of that cohort with probability one; finite cumulative exit leaves a nonzero survival probability. The cumulative effect of the full sequence therefore determines late survival.

The low-temperature limit of the thermal sector follows directly from the parent spectrum. For τ_β with $\beta > 1$, the lowest level is $m = 1$, and the total excited-state occupation and mean energy satisfy

$$\begin{aligned} e_{\mathrm{exc}}(\beta) &= 1 - \frac{1}{\zeta(\beta)} \sim 2^{-\beta}, \\ U(\beta) &\sim (\log 2) 2^{-\beta}, \\ C_\beta &\sim (\log 2)^2 2^{-\beta} \qquad (\beta \to \infty). \end{aligned}$$

(5.54)

A fixed integer cutoff including $m = 1, 2$ has the same leading terms. Excitations remain at every finite β, although the ground state can be approached to any prescribed finite precision at a positive temperature. Since the thermal state is diagonal in energy, its trace distance from the ground state is exactly

$$\frac{1}{2} \left\| \tau_\beta - |1\rangle\langle 1| \right\|_1 = e_{\mathrm{exc}}(\beta).$$

(5.55)

When this quantity is below a measurement tolerance, the event probabilities of any single measurement on this thermal sector differ by no more than that tolerance between the two states. Concentration of parent-spectrum thermal occupation does not remove other matter, auxiliaries, or saved records. The lowest-energy shell of a finite reduction may also be degenerate, retaining its internal state according to the actual preparation.

This approach can be evaluated directly. Energies in the table below are in units of E_*, and inverse temperature uses the same calibration as $T/T_* = 1/\beta$ in Chapter 3:

β	Excited-state occupation e_{exc}	Mean energy U
5	3.56127×10^{-2}	2.75562×10^{-2}

β	Excited-state occupation e_{exc}	Mean energy U
10	9.93587×10^{-4}	6.96340×10^{-4}
20	9.53961×10^{-7}	6.61352×10^{-7}

Probability tolerances 10^{-3}, 10^{-6}, and 10^{-9} correspond to inverse temperatures 9.99081, 19.93201, and 29.89736, respectively. General thermodynamic studies relate finite resources, operation time, and the attainability of zero temperature. [30] Here we determine the duration of this low-temperature process from the energy current and matter clock.

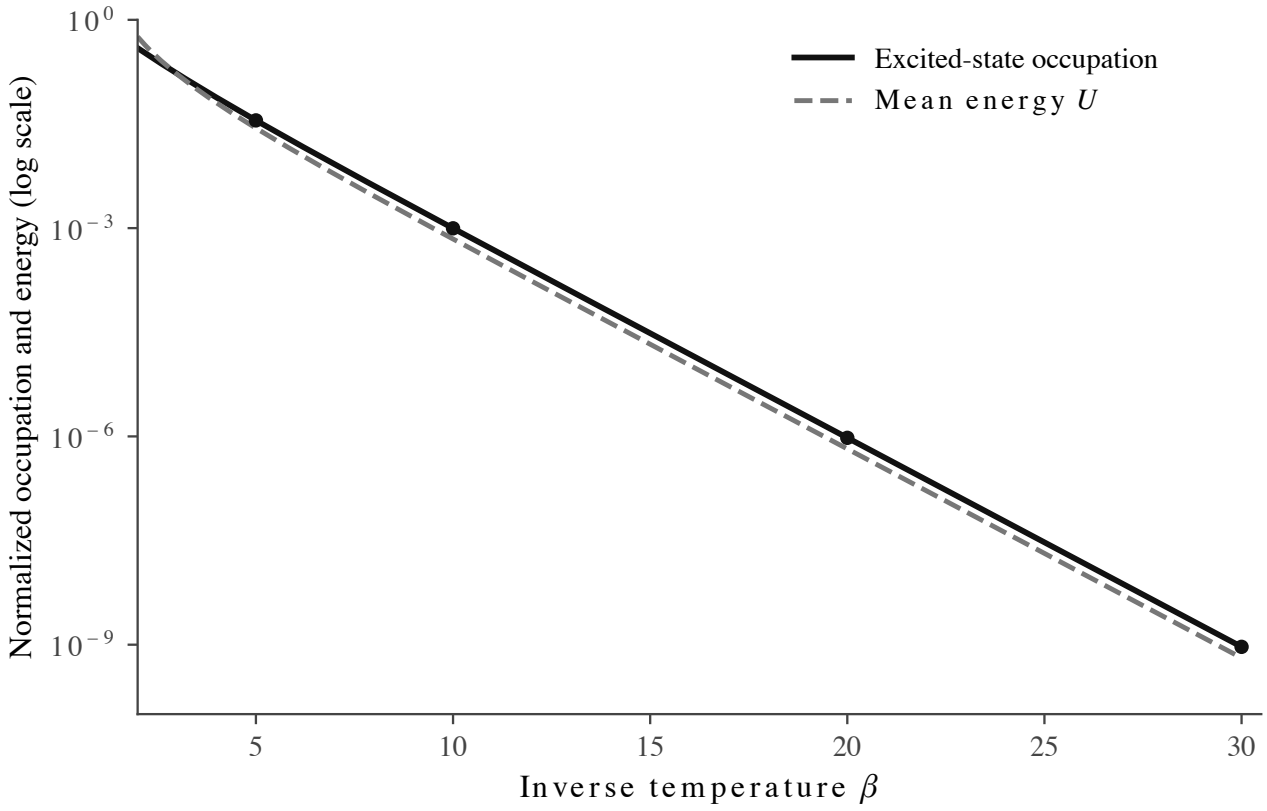


Figure 5-7. Low-temperature concentration of the parent thermal sector. Excited-state occupation and mean energy both decay with leading behavior $2^{-\beta}$, but have different normalizations. The horizontal axis is inverse temperature; the time required to reach each point remains determined by actual exchange and matter clocks.

Extending the thermal-duration relation of Section 5.1 to low temperatures, whether the exact zero-temperature boundary lies at finite matter time is determined by convergence of $\int_{\beta_0}^{\infty} F(\beta) C_\beta / J(\beta)\, d\beta$. The monotonic thermal stage and a readable matter clock must cover the entire integration range. The parent spectrum determines the decay of the thermal response, while exchange and matter phase supply J and F, respectively.

For the collective-cooling trajectory of Section 3.5, integrating $\dot{\beta}_c = U/t_*$ gives a comparison of process durations at low temperatures:

$$\frac{\Delta t_c(5 \to 10)}{t_*} = 1.95484 \times 10^3,$$
$$\frac{\Delta t_c(5 \to 20)}{t_*} = 2.17992 \times 10^6.$$

(5.56)

The closer the low-temperature endpoint, the longer it takes to reduce excitation further. These values are measured in the exchange-determined unit t_*, with finite-realization errors as in Section 3.5. Conversion to matter time inserts the clock rate F into the integral as in Section 5.1.

For nonthermal successors, the complete cycle of Chapter 3 can be tested directly. The kernel of D contains states with no net heat removal in one cycle, but residual energy persists only if subsequent operations keep them in that kernel. The largest such invariant subspace is determined by all successor conditions:

$$\mathcal{K}_\infty = \bigcap_{n \geq 0} \ker\left[(\Phi^\dagger)^n(D)\right].$$

(5.57)

Within the proven sector $\mathcal{H}_{\mathrm{sec}}$, consisting of zero execution flag, source bypass, zero momentum for both matter species, and canonical finite-store support, the two-step condition has already stabilized, giving

$$\mathcal{K}_\infty \cap \mathcal{H}_{\mathrm{sec}} = \ker\left[(D + \Phi^\dagger(D))\big|_{\mathcal{H}_{\mathrm{sec}}}\right].$$

(5.58)

Besides zero-heat states, this space contains positive-heat states blocked by finite-store boundaries from returning to the cooling port. The actual joint input has nonzero weight in this part. Every finite prefix of cycles for which records and control resources remain available therefore retains positive residual energy. This space pertains to the fixed operations and specified sector; adding the established boundary service makes some blocked branches accessible to thermal exchange again. Actual occupation and dynamical accessibility jointly determine late residual heat.

The original finite cooling program also gives a positive lower bound on active mean thermal energy. Source action may persist in the zero-bare-heat space, while a dark thermal branch may be orthogonal to the matter clock being read. Heat, coherence readability, and evolution of the whole therefore have different termination conditions.

Late scale behavior also depends on resource transport and return flows. Using the fixed participating pool of Chapter 3, write $\bar{\xi} = \langle X_{\mathrm{av}} \rangle$ and $\mathcal{C} = \langle C_{\mathrm{cont}} \rangle$, so current availability is $h = 1/2 - \bar{\xi} - \mathcal{C}$. If one cycle transfers net occupation J_{pool} into a pool of N resources, while other motion and exchanges within the pool preserve total occupation, then

$$\Delta h = -\frac{J_{\mathrm{pool}}}{N} - \Delta \mathcal{C}.$$

(5.59)

The quantity J_{pool} counts resource occupation, while J measures thermal energy current. Restoration and relative motion change matter's connection to previously written regions, while sustained net writing increases the load of the whole pool. Through the cross coherence of Section 4.4, these changes enter the clock rate. The returns and autonomous service in Section 5.3 show a decline of restoration gains and a long-term plateau, respectively; subsequent resource transport determines how this behavior continues.

Chapter 6. Conclusions

Starting from the Bost–Connes parent structure and its finite reductions, we have connected the universe as a whole, the evolution of matter and background, and internal observation. Different primes and resolution depths specify observational scopes, while joint coherence and correlations connect the physical properties within them. Matter and its environment both belong to this whole, and the state obtained by a finite observation is the manifestation of the common state in the corresponding channel.

The basic object of the full evolution is a joint state containing matter, background, energy stores, auxiliaries, and records. One operation changes these participants, and its actual successor enters the next. Let η_j^{ready} denote fresh independently prepared resources introduced in the current cycle, with old participants and their records retained in Ω_j. The common process and record probabilities are

$$\boxed{\begin{aligned}\Omega_{j+1} &= \mathbb{U}_j(\Omega_j \otimes \eta_j^{\mathrm{ready}})\mathbb{U}_j^{\dagger},\\ q_\alpha &= \mathrm{Tr}(\Omega_{j+1}R_\alpha).\end{aligned}} \tag{6.1}$$

Formation, maintenance, and exit of matter, background response, and observed events can thus be calculated together from actual operations. The used source state continues to affect restoration, which changes subsequent maintenance and exchange. Cosmic evolution therefore has a memory carried by the joint state.

Heat exchange is energy transfer within this evolution. We have given an energy-conserving connection between thermal reception and resource restoration, compensating the bare thermal gap and matter maintenance energy separately while retaining positive and negative matter branches and conditional store correlations. For a time-independent total Hamiltonian, sector energies and interaction energy are jointly conserved. When the thermal sector follows the parent thermal-state family, net outgoing power normalized by E_* further determines the advance of inverse temperature:

$$\boxed{\begin{aligned}&\textstyle\sum_A J_A + \frac{d}{dt}\langle V\rangle = 0,\\ &J = C_\beta \dot{\beta}, \qquad C_\beta = \mathrm{Var}_{\tau_\beta}(H_{\log}).\end{aligned}} \tag{6.2}$$

Collective exchange and its internal program give a comparison trajectory $\dot{\beta}_c = U/t_*$ with controlled error, where t_* includes exchange, transfer to the store, and execution speed. The nonthermal loaded cycle gives per-cycle heat removal and cumulative balances. The two realizations connect energy transfer to temperature evolution and resource service, respectively.

Matter also provides observational references in this process. Its phase is determined by the cross coherence of the loaded matter, with source back-action and background correlations entering the clock rate together. Emission, propagation, and reception connect distant events. When several matter clocks and light propagation in different directions are compatible over the same range, full time accounting gives

$$\boxed{\begin{aligned} d\tau &= F(t)\,dt, \\ a(t) &= \frac{F(t)/G(t)}{F(t_0)/G(t_0)}, \\ H_{\mathrm{geom}} &= \frac{1}{F}\left(\frac{\dot{F}}{F} - \frac{\dot{G}}{G}\right). \end{aligned}} \tag{6.3}$$

Scale change depends on the relative responses of matter phase and light propagation. Restoration gives finite growth and slowly varying spectral-line redshift. Early emission retaining environmental correlations also produces nonzero, distinguishable late reception records. Long-term service further reveals a process for preserving bias: individual clock multipliers can remain above their input values and approach a common plateau, while instantaneous growth still undergoes reversals.

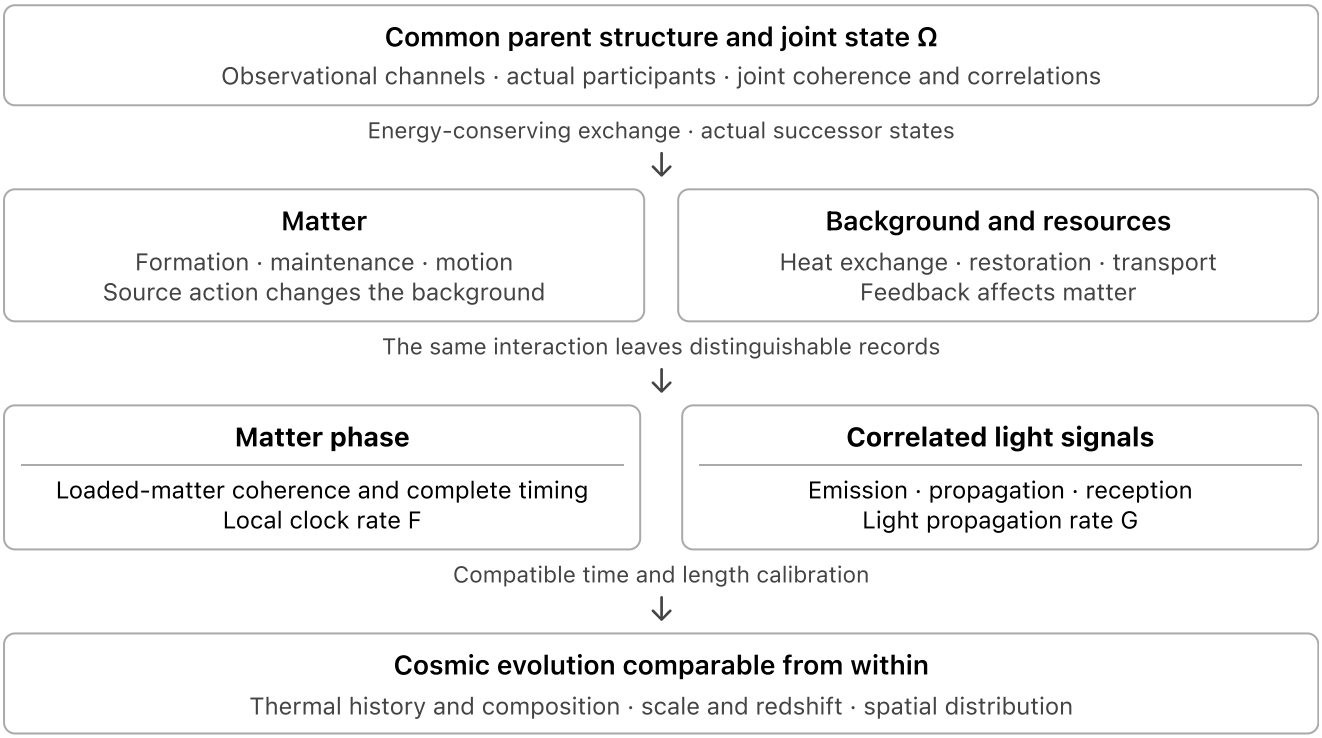


Figure 6-1. Common relations between the universe as a whole, its evolution, and internal observation. Matter and background change one another within the joint state, passing energy and correlations to subsequent states. Records, matter clocks, and light signals turn these changes into comparable events and scales.

This line of reasoning also organizes early- and late-universe phenomena. The critical parent spectrum broadens energy participation, formation and survival determine matter composition, and matter distribution produces gravitational feedback through background response. With the source transfer, return, and physical normalization used here, the net baryon ratio is $\eta_\star \simeq 6.19 \times 10^{-10}$, and the dark-component-to-baryon density ratio is 16/3. At late times, the fixed cycle retains positive residual energy in an occupied dark thermal subspace, while boundary service can release part of it. Long-term thermal evolution is thus determined jointly by state occupation and dynamical accessibility.

Heat exchange and restoration change matter and background; matter phases and correlated light signals turn these changes into comparable readings; resource supply and dynamical accessibility determine how long the processes can continue. Connecting these mechanisms within the same preparation, dynamics, and time accounting is the common direction for establishing a complete scale history, a distance–redshift relation, and observational comparisons of thermal history.

References

[1] Shao-Wu Hu. *Dynamics of the Bost–Connes State Space*. Version 2. Zenodo (2026). doi:10.5281/zenodo.23005935.

[2] Shao-Wu Hu. *The Unification of Matter, Forces, and Spacetime from the Bost–Connes System*. Version 7. Zenodo (2026). doi:10.5281/zenodo.23392111.

[3] Jean-Benoît Bost and Alain Connes. *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*. Selecta Mathematica, New Series **1**, 411–457 (1995). doi:10.1007/BF01589495.

[4] Alain Connes and Matilde Marcolli. *Noncommutative Geometry, Quantum Fields and Motives*. American Mathematical Society (2008), Theorem 3.32. Author manuscript.

[5] Marcelo Laca and Iain Raeburn. *A Semigroup Crossed Product Arising in Number Theory*. Journal of the London Mathematical Society **59**, 330–344 (1999). doi:10.1112/S0024610798006620.

[6] Matteo Lostaglio, David Jennings and Terry Rudolph. *Description of quantum coherence in thermodynamic processes requires constraints beyond free energy*. Nature Communications **6**, 6383 (2015). doi:10.1038/ncomms7383.

[7] Elliott H. Lieb and Mary Beth Ruskai. *Proof of the strong subadditivity of quantum-mechanical entropy*. Journal of Mathematical Physics **14**, 1938–1941 (1973). doi:10.1063/1.1666274.

[8] Wiesław Pusz and Stanisław L. Woronowicz. *Passive states and KMS states for general quantum systems*. Communications in Mathematical Physics **58**, 273–290 (1978). doi:10.1007/BF01614224.

[9] Fernando Brandão, Michał Horodecki, Nelly Ng, Jonathan Oppenheim and Stephanie Wehner. *The second laws of quantum thermodynamics*. Proceedings of the National Academy of Sciences **112**, 3275–3279 (2015). doi:10.1073/pnas.1411728112.

[10] David Reeb and Michael M. Wolf. *An improved Landauer principle with finite-size corrections*. New Journal of Physics **16**, 103011 (2014). doi:10.1088/1367-2630/16/10/103011.

[11] Massimiliano Esposito, Katja Lindenberg and Christian Van den Broeck. *Entropy production as correlation between system and reservoir*. New Journal of Physics **12**, 013013 (2010). doi:10.1088/1367-2630/12/1/013013.

[12] Wojciech H. Zurek. *Quantum Darwinism*. Nature Physics **5**, 181–188 (2009). doi:10.1038/nphys1202.

[13] W. Forrest Stinespring. *Positive functions on C*-algebras*. Proceedings of the American Mathematical Society **6**, 211–216 (1955). doi:10.1090/S0002-9939-1955-0069403-4.

[14] E. B. Davies and J. T. Lewis. *An operational approach to quantum probability*. Communications in Mathematical Physics **17**, 239–260 (1970). doi:10.1007/BF01647093.

[15] Masanao Ozawa. *Quantum measuring processes of continuous observables*. Journal of Mathematical Physics **25**, 79–87 (1984). doi:10.1063/1.526000.

[16] H. Salecker and E. P. Wigner. *Quantum Limitations of the Measurement of Space-Time Distances*. Physical Review **109**, 571–577 (1958). doi:10.1103/PhysRev.109.571.

[17] Paul Erker, Mark T. Mitchison, Ralph Silva, Mischa P. Woods, Nicolas Brunner and Marcus Huber. *Autonomous Quantum Clocks: Does Thermodynamics Limit Our Ability to Measure Time?* Physical Review X **7**, 031022 (2017). doi:10.1103/PhysRevX.7.031022.

[18] R. V. Pound and G. A. Rebka, Jr. *Apparent Weight of Photons*. Physical Review Letters **4**, 337–341 (1960). doi:10.1103/PhysRevLett.4.337.

[19] Roy J. Glauber. *The Quantum Theory of Optical Coherence*. Physical Review **130**, 2529–2539 (1963). doi:10.1103/PhysRev.130.2529.

[20] P. L. Kelley and W. H. Kleiner. *Theory of Electromagnetic Field Measurement and Photoelectron Counting*. Physical Review **136**, A316–A334 (1964). doi:10.1103/PhysRev.136.A316.

[21] Planck Collaboration, N. Aghanim et al. *Planck 2018 results. VI. Cosmological parameters*. Astronomy & Astrophysics **641**, A6 (2020). doi:10.1051/0004-6361/201833910.

[22] D. J. Fixsen. *The Temperature of the Cosmic Microwave Background*. The Astrophysical Journal **707**, 916–920 (2009). doi:10.1088/0004-637X/707/2/916.

[23] Rudolf Haag, Nico M. Hugenholtz and Marinus Winnink. *On the equilibrium states in quantum statistical mechanics*. Communications in Mathematical Physics **5**, 215–236 (1967). doi:10.1007/BF01646342.

[24] Shao-Wu Hu. *The Unification of Matter, Forces, and Spacetime from the Bost–Connes System*. Version 6. Zenodo (2026). doi:10.5281/zenodo.22365347.

[25] Jeffrey A. Harvey and Michael S. Turner. *Cosmological baryon and lepton number in the presence of electroweak fermion-number violation*. Physical Review D **42**, 3344–3349 (1990). doi:10.1103/PhysRevD.42.3344.

[26] Ryan J. Cooke, Max Pettini and Charles C. Steidel. *One Percent Determination of the Primordial Deuterium Abundance*. The Astrophysical Journal **855**, 102 (2018). doi:10.3847/1538-4357/aaab53.

[27] Adam G. Riess et al. *Observational Evidence from Supernovae for an Accelerating Universe and a Cosmological Constant*. The Astronomical Journal **116**, 1009–1038 (1998). doi:10.1086/300499.

[28] M. Abdul Karim et al. (DESI Collaboration). *DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints*. Physical Review D **112**, 083515 (2025). doi:10.1103/tr6y-kpc6.

[29] Douglas Clowe, Maruša Bradač, Anthony H. Gonzalez, Maxim Markevitch, Scott W. Randall, Christine Jones and Dennis Zaritsky. *A Direct Empirical Proof of the Existence of Dark Matter*. The Astrophysical Journal **648**, L109–L113 (2006). doi:10.1086/508162.

[30] Lluís Masanes and Jonathan Oppenheim. *A general derivation and quantification of the third law of thermodynamics*. Nature Communications **8**, 14538 (2017). doi:10.1038/ncomms14538.